



Chapter 9 IIR Digital Filter Design

- *Preliminary Considerations*
- *IIR Digital Filter Design Methods*
- *MATLAB Functions in IIR Filter Design*



Filter Design Objective



- Objective

Determination of a realizable transfer function $G(z)$ approximating a given frequency response specification is an important step in the development of a digital filter.

- If an IIR filter is desired, $G(z)$ should be a **stable real rational** function.
- Digital filter design is the process of deriving the transfer function $G(z)$.

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9.1 Preliminary Considerations



- These are two issues that need to be considered before one can develop the digital transfer function $G(z)$.
 1. The first and foremost issue is the development of a reasonable filter frequency response specification.
 2. The second issue is to determine whether an FIR or an IIR digital filter is to be designed.

9.1.1 Digital Filter Specifications



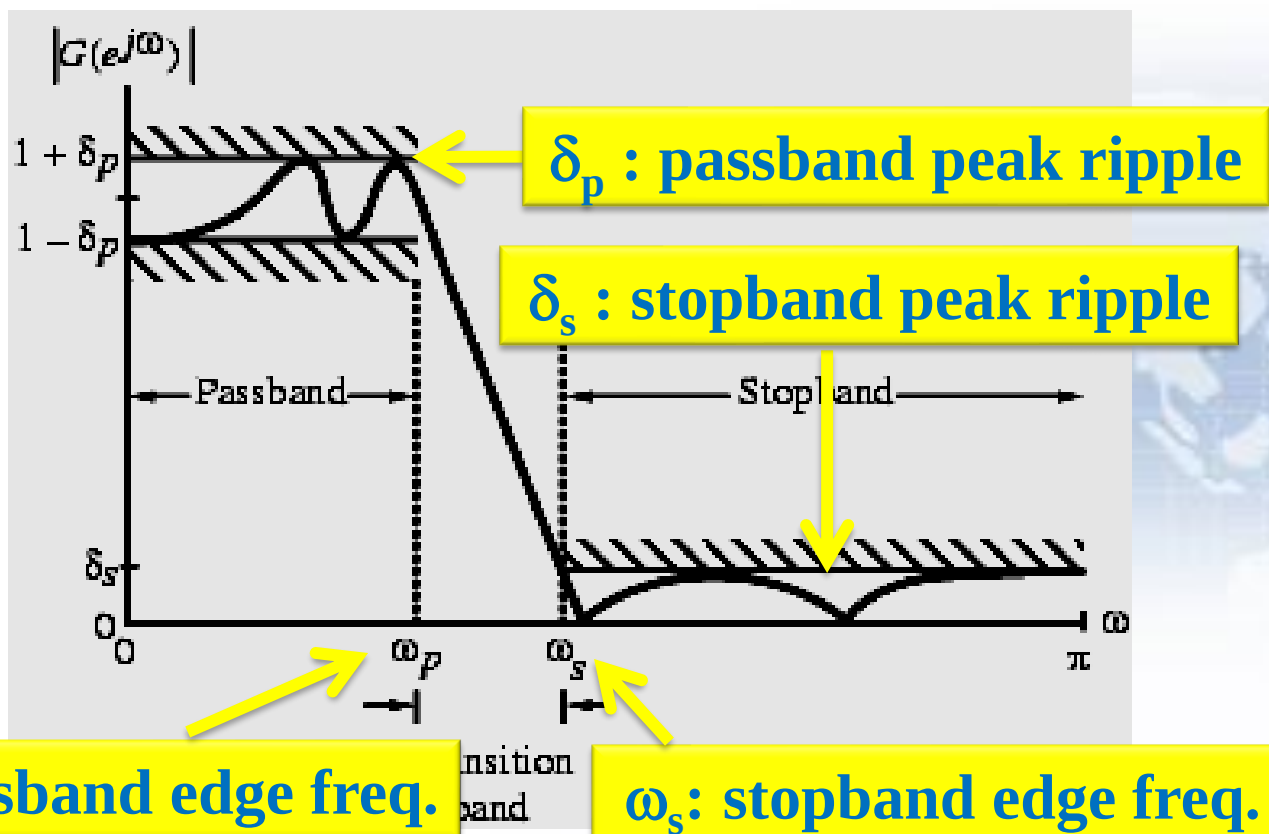
- Usually, either the **magnitude and/or the phase response** is specified for the design of digital filter for most applications;
- In most practical applications, the problem of interest is the development of a realizable approximation to a given **magnitude response specification**.
- We restrict to the **magnitude approximation problem** in this chapter.

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Digital Filter Design Specifications



- The magnitude response $|G(e^{j\omega})|$ of a digital lowpass filter may be specified below:



In the passband, we require
 $1 - \delta_p \leq |G(e^{j\omega})| \leq 1 + \delta_p$,

In the stopband, we require
 $|G(e^{j\omega})| \leq \delta_s$

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Digital Filter Design Specifications



- Specifications are often given in terms of loss function:

$$A(\omega) = -20 \log_{10} |G(e^{j\omega})| \text{ dB}$$

- Peak passband ripple:

$$\alpha_p = -20 \log_{10} (1 - \delta_p) \text{ dB}$$

E.g., $\alpha_p = 1 \text{ dB} \rightarrow \delta_p = 0.1087$

- Minimum stopband attenuation:

$$\alpha_s = -20 \log_{10} (\delta_s) \text{ dB}$$

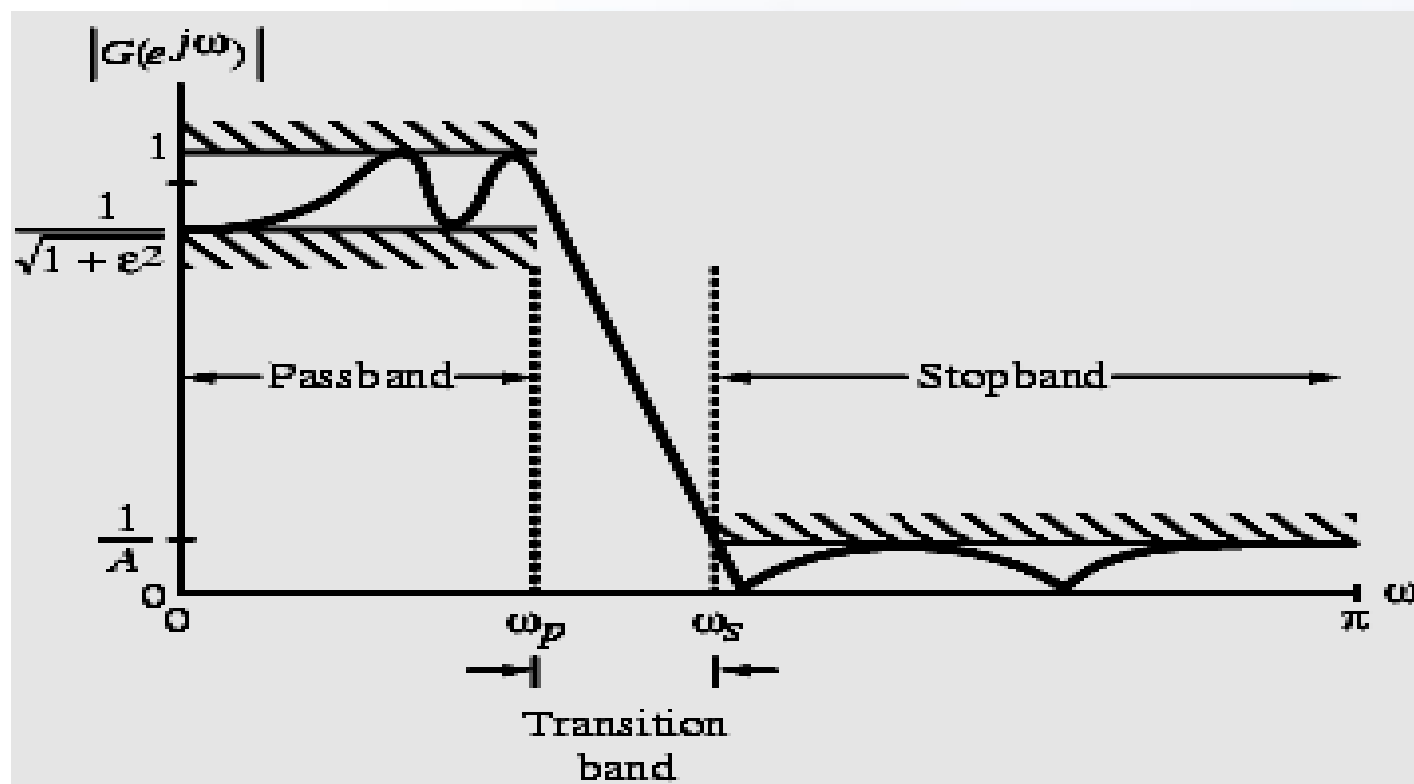
E.g., $\alpha_s = 80 \text{ dB} \rightarrow \delta_s = 0.0001$

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9.1.1 Digital Filter Specifications



Magnitude specifications may alternately be given in a **normalized form** as indicated below:



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9.1.1 Digital Filter Specifications



- Here, the maximum value of the magnitude in the passband is assumed to be **unity**.
 - $1 / \sqrt{1 + \varepsilon^2}$ - Maximum passband deviation, given by the minimum value of the magnitude in the passband.
 - $1/A$ - Maximum stopband magnitude;
Minimum stopband attenuation is: $20\log_{10}(A)$ dB.

9.1.1 Digital Filter Specifications



- For the normalized specification, maximum value of the gain function or the minimum value of the loss function is 0 dB.
- Maximum passband attenuation:

$$\alpha_{\max} = 20 \log_{10} \left(\sqrt{1 + \varepsilon^2} \right) \text{ dB}$$

- For $\delta_p \ll 1$, it can be shown that:

$$\alpha_{\max} = -20 \log_{10} (1 - 2 \delta_p) \text{ dB}$$

Digital Filter Design Specifications

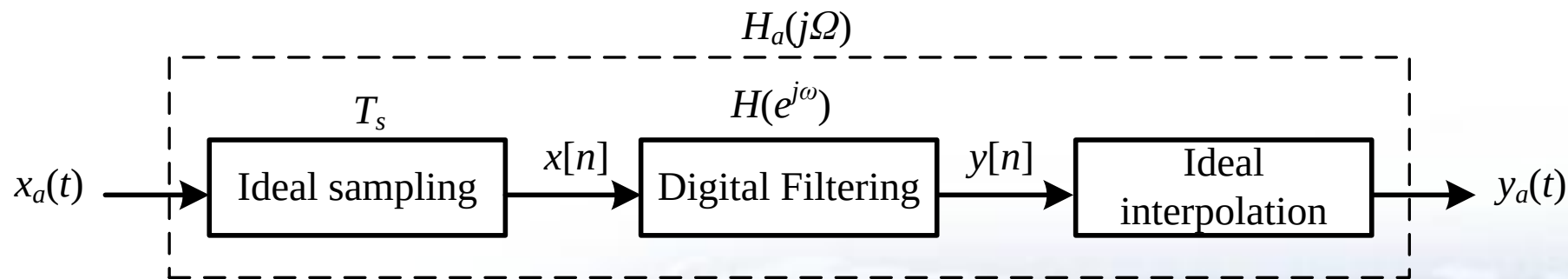


Fig. Ideal data flow for digital processing of continuous signal

- In practice, passband edge frequency f_p and stopband edge frequency f_s are specified in **Hz**.
- For digital filter design, normalized angular edge frequencies can be computed in radians as

$$\omega_p = \frac{2\pi f_p}{F_s} = 2\pi f_p T_s \quad \omega_s = \frac{2\pi f_s}{F_s} = 2\pi f_s T_s$$

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where F_s and T_s are sampling frequency and sampling interval

9.1.1 Digital Filter Specifications



- Example 9.1

Let $\alpha_p=0.1\text{dB}$, and $\alpha_s= 35\text{dB}$. Determine the δ_p and δ_s

$$\delta_p = 1 - 10^{-\alpha_p/20} = 1 - 10^{-0.005} = 0.0114469$$

$$\delta_s = 10^{-\alpha_s/20} = 10^{-1.75} = 0.01778279$$

9.1.1 Digital Filter Specifications



- Example 9.2

Let $F_p = 7$ kHz, $F_s = 3$ kHz, and $F_T = 25$ kHz

- Then

$$\omega_p = \frac{2\pi(7 \times 10^3)}{25 \times 10^3} = 0.56\pi$$

$$\omega_s = \frac{2\pi(3 \times 10^3)}{25 \times 10^3} = 0.24\pi$$

9.1.2 Selection of the Filter Type



- The *objective* of digital filter design is to develop a **casual, stable transfer function $G(z)$** which meets the FR specifications.
- Whether IIR or FIR should be selected?
 - **IIR**: lower order to reduce computational complexity; but must be stable.
 - **FIR**: linear phase; always stable; but higher order.

9.1.3 Basic Approach to IIR Digital Filter Design



- Most common approach to IIR filter design:

Convert the digital filter specifications into **analog lowpass** prototype one.



Determine the **analog lowpass** filter $H_a(s)$ meeting these specifications.



Transform it into the desired **digital** filter transfer function $G(z)$.

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9.1.3 Basic Approach to IIR Digital Filter Design



- Why this approach has been widely used:
 - (1) Analog approximation techniques are highly advanced.
 - (2) They usually yield closed-form solutions.
 - (3) Extensive tables are available for analog filter design.
 - (4) Many applications require digital simulation of analog systems.

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9.1.3 Basic Approach to IIR Digital Filter Design



- The basic idea is to apply a mapping from **s-domain** to the **z-domain** so that the essential properties of the analog frequency response are preserved. [$H_a(s) \rightarrow G(z)$]
- This implies that the mapping should be:
 - (1) The imaginary axis in the s-plane be mapped onto the unit circle of the z-plane.
 - (2) A stable analog transfer function be transformed into a stable digital transfer function.

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9.1.4 IIR Digital Filter Order Estimation



- After the type of the digital filter has been selected, the next step in the filter design process is to estimate the filter order N .
- For the design of **IIR LPF**, the order of $H_a(s)$ is estimated from its specifications using the appropriate formula given in Eq.(A.9), (A.17), or (A.27), depending on which approximation is desired.
- Then the order of $G(z)$ can be determined automatically from the transformation being used to convert $H_a(s)$ into $G(z)$.

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9.1.5 Scaling the Digital Transfer function



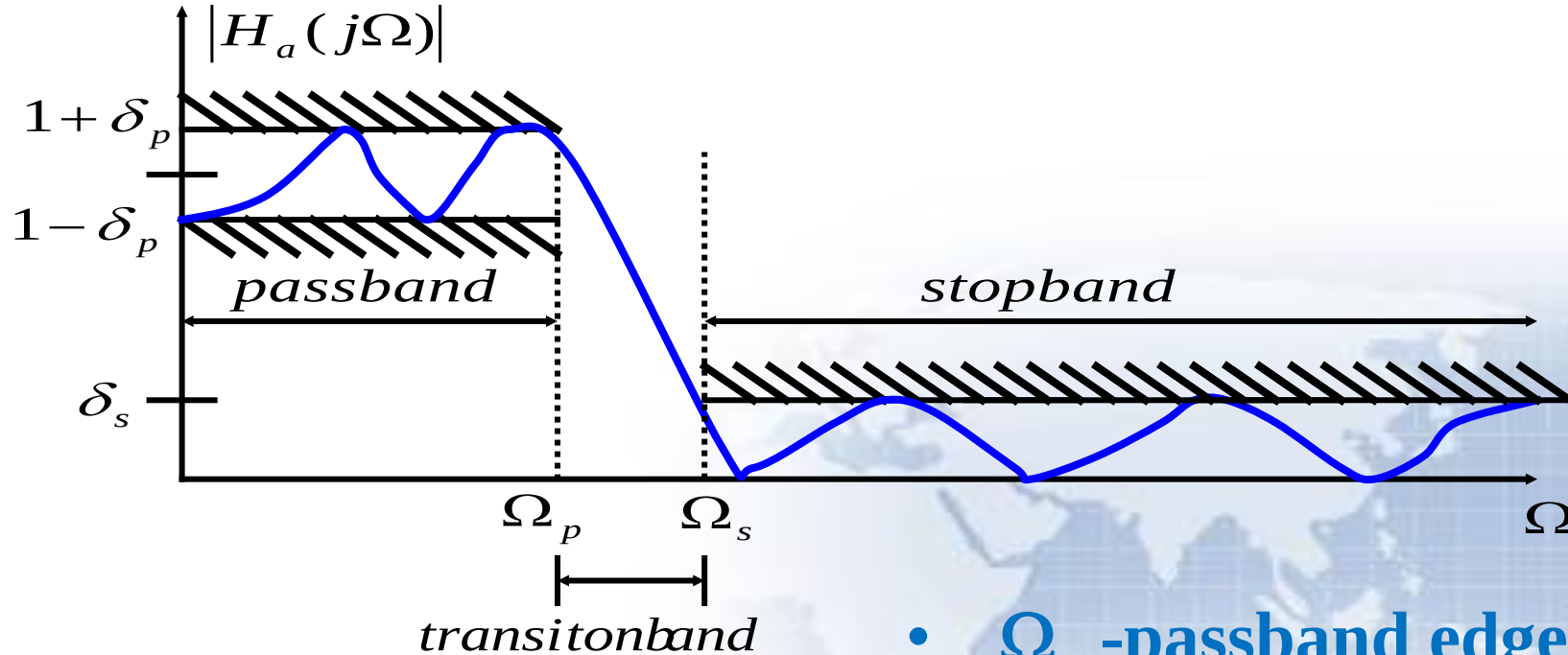
- In magnitude scaling, the transfer function $G(z)$ is multiplied by a **constant K** so that the maximum magnitude of the $KG(z)$ in the passband is unity; it has a maximum gain of **0dB**.
- If G_{max} is the maximum value of $|G(e^{j\omega})|$ in the passband, then $K=1/G_{max}$
 - For a lowpass $G(z)$ with a maximum magnitude at DC, $K=1/G(1)$;
 - For a highpass $G(z)$ with a maximum magnitude at $\omega=\pi$, $K=1/G(-1)$;
 - For a bandpass $G(z)$, $K=1/Ge^{j\omega_c}$, where ω_c is the center frequency of the passband.

Appendix A: Analog Filter Design



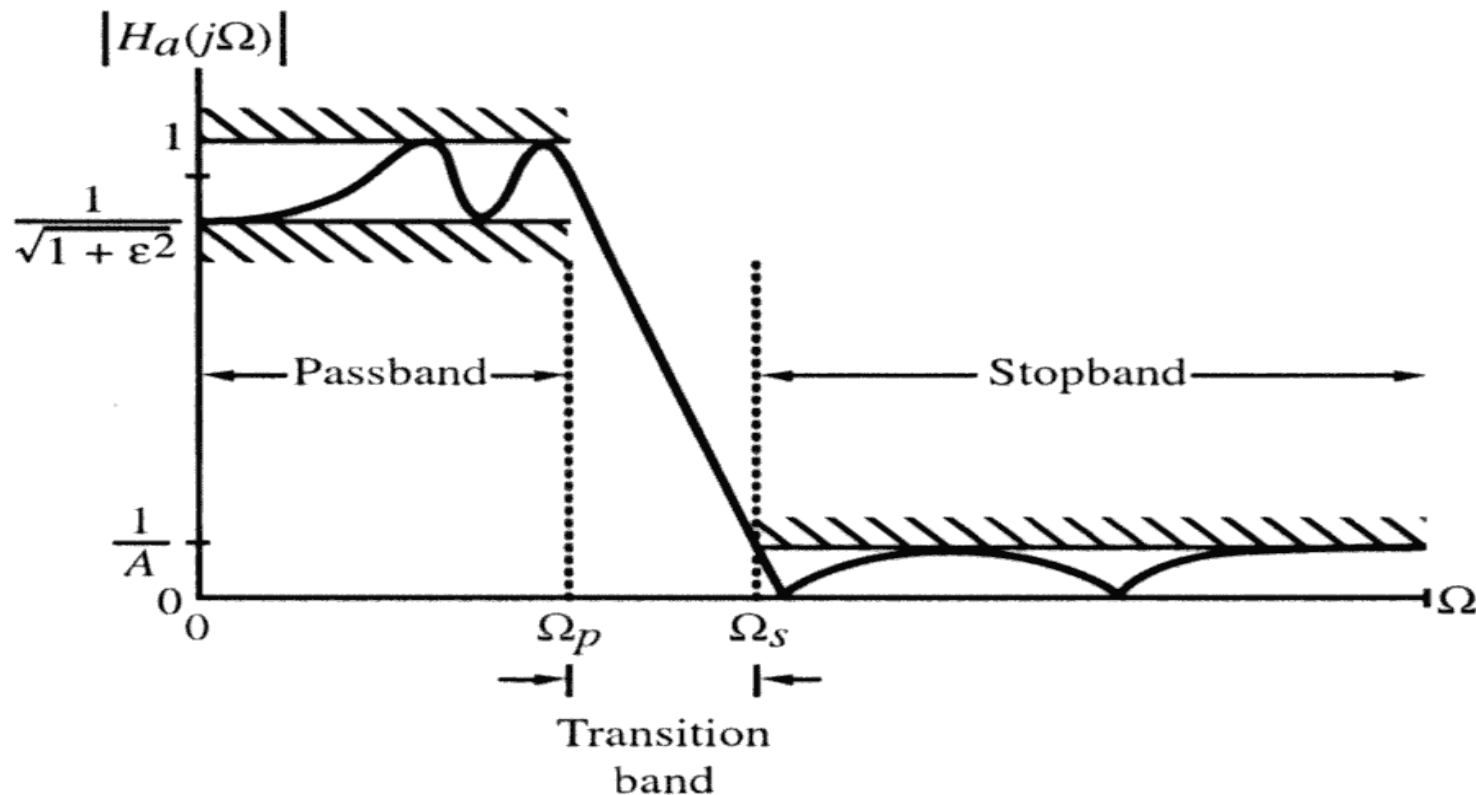
- **Analog Filter Specifications**
- **Butterworth approximation**
- **Chebyshev approximation**
 Type1 Type2
- **Elliptic approximation**
- **Analog Lowpass Filter Design Using MATLAB**

Analog Lowpass Filter Specifications



- Ω_p -passband edge frequency
- Ω_s -stopband edge frequency
- δ_p -peak ripple value in the passband
- δ_s -peak ripple value in the stopband

Normalized magnitude specifications for an analog lowpass filter



Transition ratio or selectivity parameter:

$$K = \Omega_p / \Omega_s$$

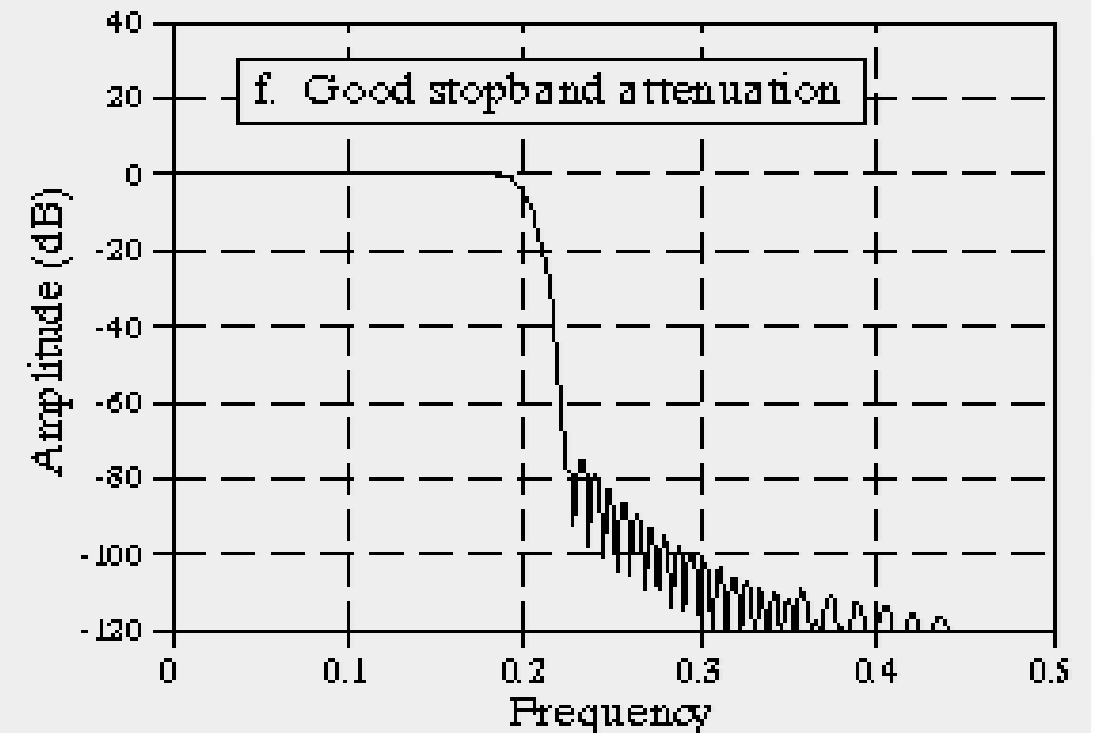
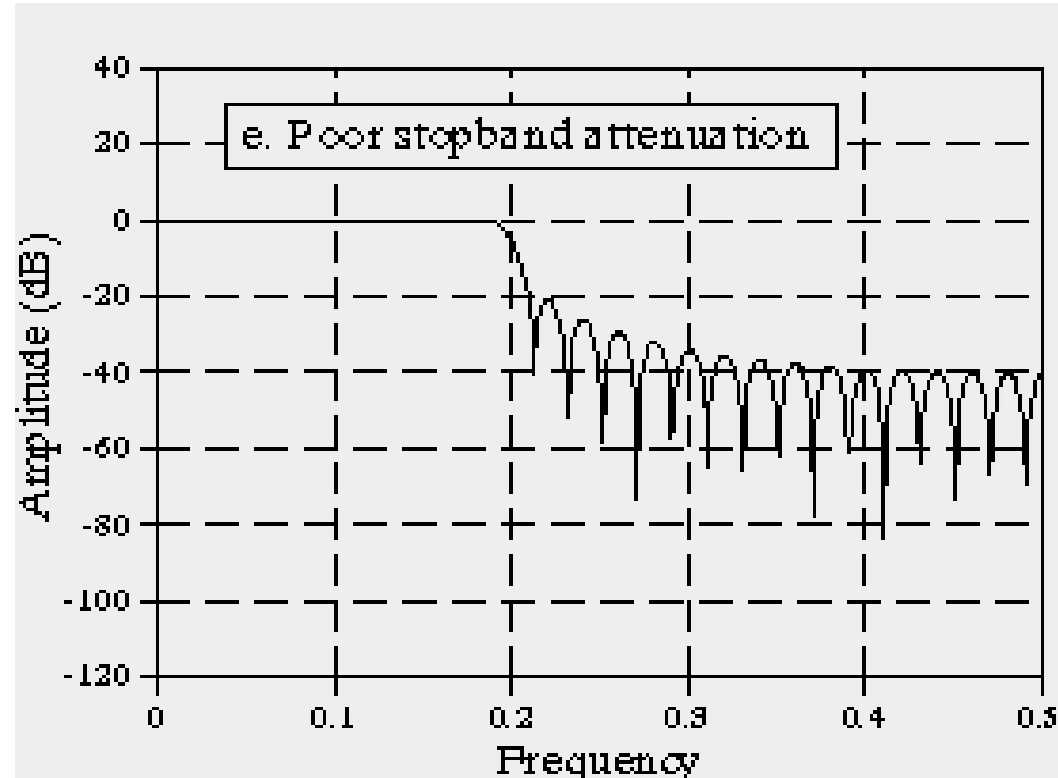
Discrimination parameter:

$$K_1 = \epsilon / \sqrt{A^2 - 1}$$



- In the **passband**, defined by $0 \leq \Omega \leq \Omega_p$
$$1 - \delta_p \leq |H_a(j\Omega)| \leq 1 + \delta_p, \quad |\Omega| \leq \Omega_p$$
- In the **stopband**, defined by $\Omega_s \leq \Omega \leq \infty$,
$$|H_a(j\Omega)| \leq \delta_s, \quad \Omega_s \leq \Omega \leq \infty$$
- **Peak passband ripple**
$$\alpha_p = -20 \log_{10}(1 - \delta_p) \text{ dB}$$
- **Minimum stopband attenuation**
$$\alpha_s = -20 \log_{10}(\delta_s) = 20 \log_{10}(A) \text{ dB}$$

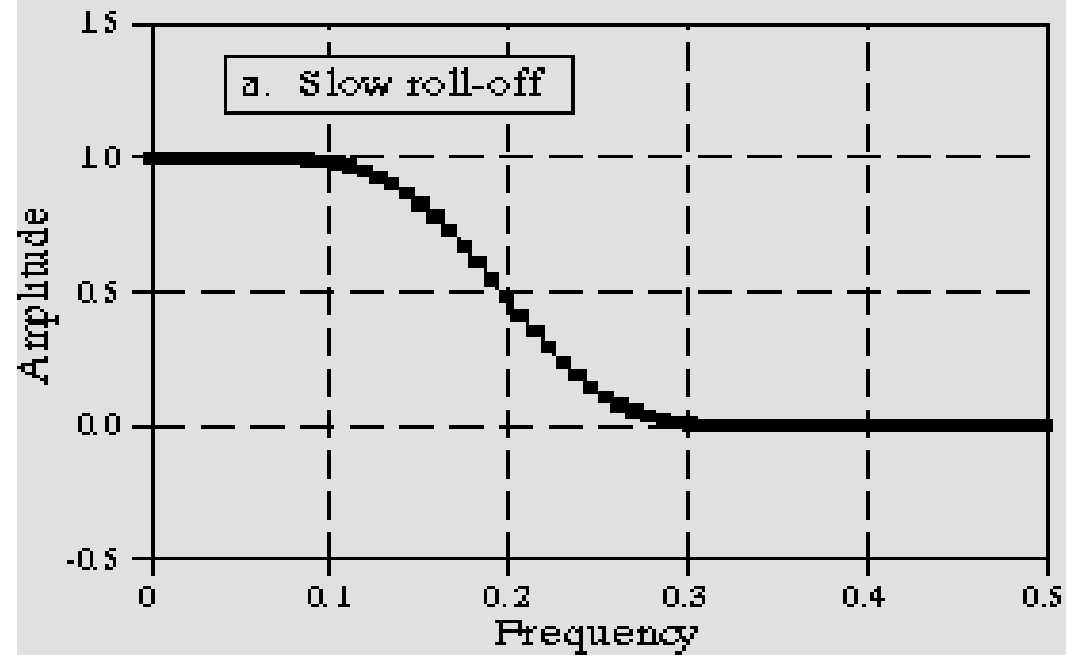
Attenuation function or Loss function



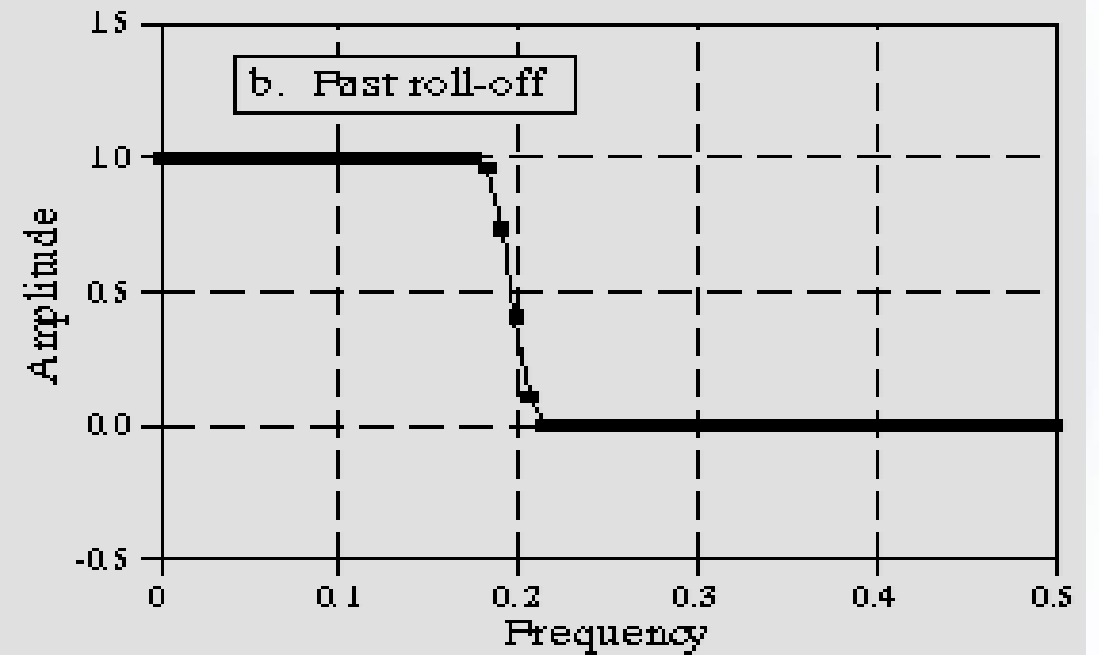
Transition ratio or Selectivity parameter



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Appendix A: Analog Filter Design



- **Butterworth Approximation**

The magnitude-square response of an N -th order analog lowpass Butterworth filter is given by:

$$|H_a(j\Omega)|^2 = \frac{1}{1 + (\Omega / \Omega_c)^{2N}}$$

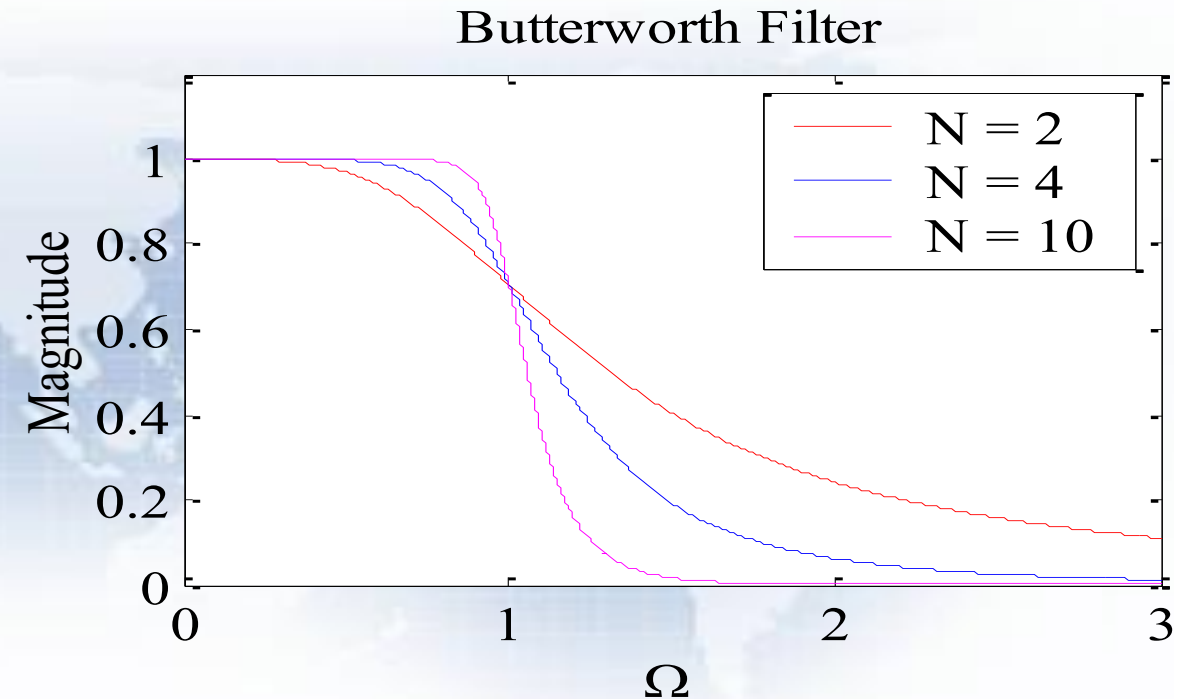
The Butterworth lowpass filter thus is said to have a maximally-flat magnitude at $\Omega = 0$.

Butterworth Approximation



- Gain in dB is:
$$G(\Omega) = 10 \log_{10} |H_a(j\Omega)|^2$$

As $G(0) = 0$ and
$$G(\Omega_c) = 10 \log_{10}(0.5) \cong -3 \text{ dB}$$
- Ω_c is called **3-dB cutoff frequency**.
- Typical magnitude responses with $\Omega_c = 1$ are shown in right.



Butterworth Approximation



- Two parameters completely characterizing a Butterworth lowpass filter are Ω_c and N .
- These are determined from the specified bandedges Ω_p and Ω_s , and minimum passband magnitude $1/\sqrt{(1+\varepsilon^2)}$, and maximum stopband ripple $1/A$.

Butterworth Approximation



- Ω_c and N are thus determined from:

$$|H_a(j\Omega_p)|^2 = \frac{1}{1 + (\Omega_p / \Omega_c)^{2N}} = \frac{1}{1 + \varepsilon^2}$$

$$|H_a(j\Omega_s)|^2 = \frac{1}{1 + (\Omega_s / \Omega_c)^{2N}} = \frac{1}{A^2}$$

Solving the above, we get:

$$N = \frac{1}{2} \cdot \frac{\log_{10}[(A^2 - 1) / \varepsilon^2]}{\log_{10}(\Omega_s / \Omega_p)} = \frac{\log_{10}(1 / k_1)}{\log_{10}(1 / k)}$$



- Transfer function of an analog Butterworth lowpass filter is given by:

$$H_a(s) = \frac{C}{D_N(s)} = \frac{\Omega_c^N}{s^N + \sum_{l=0}^{N-1} d_l s^l} = \frac{\Omega_c^N}{\prod_{l=1}^N (s - p_l)}$$

Where, $p_l = \Omega_c e^{j[\pi(N+2l-1)/2N]}$, $1 \leq l \leq N$

Butterworth Polynomial :

$D_N(S)$					
N	d_0	d_1	d_2	d_3	d_4
1	1	1			
2	1	1.4142136	1		
3	1	2.0000000	2.0000000	1	
4	1	2.6131259	3.4142136	2.6131259	1



Example

Determine the lowest order of Butterworth LPF with a 1-dB cutoff frequency at 1kHz and minimum attenuation of 40dB at 5kHz.

$$10\log_{10}\left(\frac{1}{1+\varepsilon^2}\right) = -1(dB) \quad \Rightarrow \varepsilon^2 \approx 0.25895$$

$$10\log_{10}\left(\frac{1}{A^2}\right) = -40(dB) \quad \Rightarrow A^2 = 10000$$

$$\frac{1}{k_1} = \frac{\sqrt{A^2 - 1}}{\varepsilon} = 196.51334 \quad \frac{1}{k} = \Omega_s / \Omega_p = 5$$



So we get:

$$N = \frac{\log_{10}(1/k_1)}{\log_{10}(1/k)} = \frac{\log_{10}(196.51334)}{\log_{10}(5)} = 3.2811022$$

The nearest integer N=4.

From the Butterworth Polynomial:

$$H_a(s) = \frac{1}{S^4 + 2.613S^3 + 3.414S^2 + 2.613S + 1}$$

Analog Filter Design



- **Chebyshev Approximation**

The magnitude-square response of an N -th order analog lowpass **Type 1 Chebyshev filter** is given by:

$$|H_a(s)|^2 = \frac{1}{1 + \varepsilon^2 T_N^2(\Omega / \Omega_p)}$$

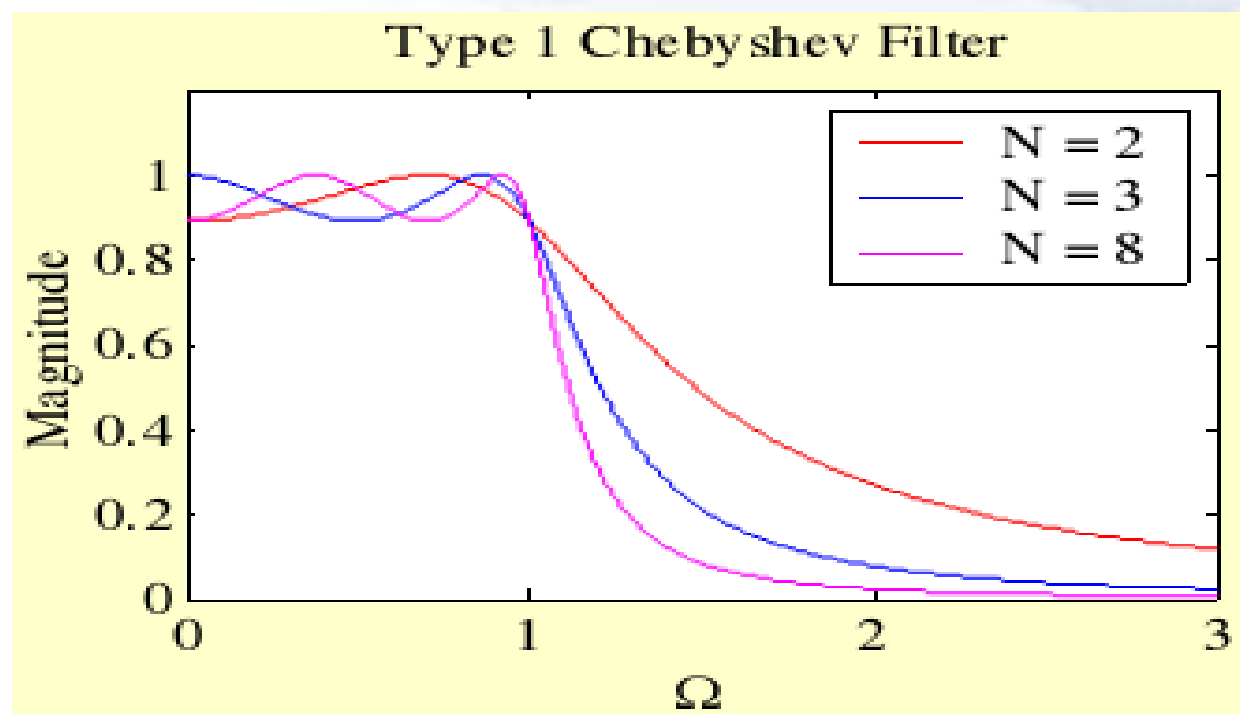
where $T_N(\Omega)$ is the **Chebyshev polynomial of order N** :

$$T_N(\Omega) = \begin{cases} \cos(N \cos^{-1} \Omega), & |\Omega| \leq 1 \\ \cosh(N \cosh^{-1} \Omega), & |\Omega| > 1 \end{cases}$$

Chebyshev Approximation



Typical magnitude response plots of the analog lowpass Type 1 Chebyshev filter are shown below:



Chebyshev Approximation



If at $\Omega = \Omega_s$ the magnitude is equal to $1/A$, then

$$|H_a(j\Omega_s)|^2 = \frac{1}{1 + \varepsilon^2 T_N^2(\Omega_s / \Omega_p)} = \frac{1}{A^2}$$

Solving the above we get:

$$N = \frac{\cosh^{-1}(\sqrt{A^2 - 1} / \varepsilon)}{\cosh^{-1}(\Omega_s / \Omega_p)} = \frac{\cosh^{-1}(1 / k_1)}{\cosh^{-1}(1 / k)}$$

- **Order N is chosen as the nearest integer greater than or equal to the above value.**

Chebyshev Approximation



The magnitude-square response of an N -th order analog lowpass **Type 2 Chebyshev** (also called **inverse Chebyshev**) filter is given by:

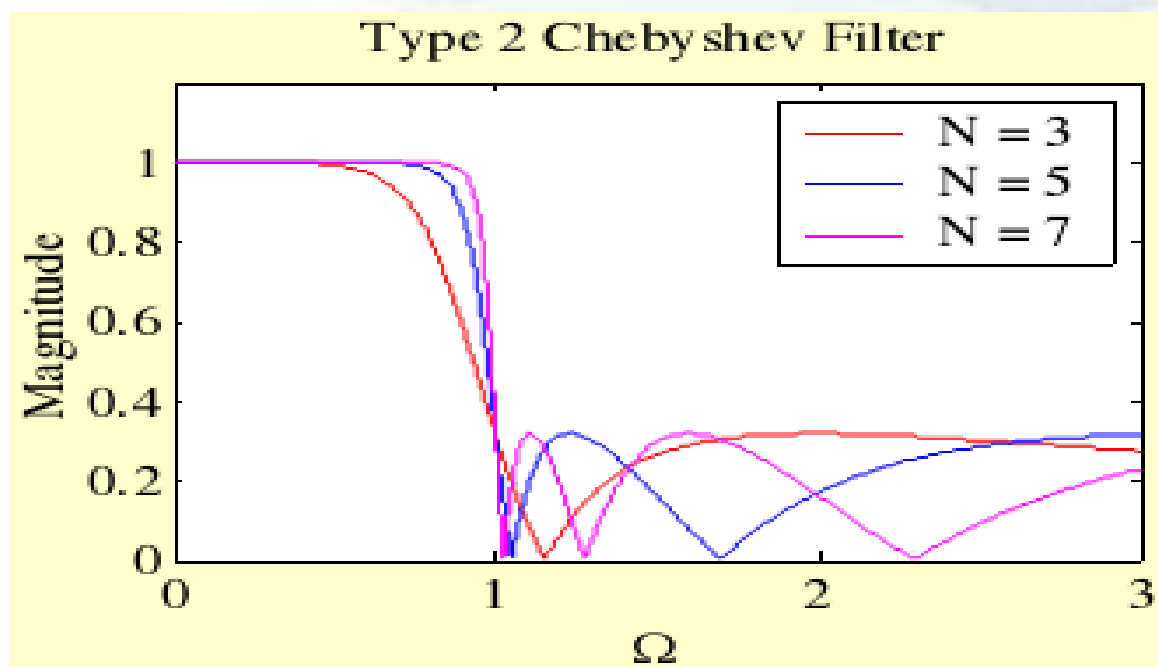
$$|H_a(j\Omega)|^2 = \frac{1}{1 + \varepsilon^2 \left[\frac{T_N(\Omega_s / \Omega_p)}{T_N(\Omega_s / \Omega)} \right]^2}$$

where $T_N(\Omega)$ is the Chebyshev polynomial of order N .

Chebyshev Approximation



- Typical magnitude response plots of the analog lowpass Type 2 Chebyshev filter are shown below:



Chebyshev Approximation



- The order N of the Type 2 Chebyshev filter is determined from given ε , Ω_s , and A using:

$$N = \frac{\cosh^{-1}(\sqrt{A^2 - 1} / \varepsilon)}{\cosh^{-1}(\Omega_s / \Omega_p)} = \frac{\cosh^{-1}(1 / k_1)}{\cosh^{-1}(1 / k)}$$

- Example - Determine the lowest order of a Chebyshev lowpass filter with a 1-dB cutoff frequency at 1kHz and a minimum attenuation of 40 dB at 5kHz.

$$N = \frac{\cosh^{-1}(1 / k_1)}{\cosh^{-1}(1 / k)} = 2.60591$$

Analog Filter Design



- **Elliptic Approximation**

The square-magnitude response of an elliptic lowpass filter is given by:

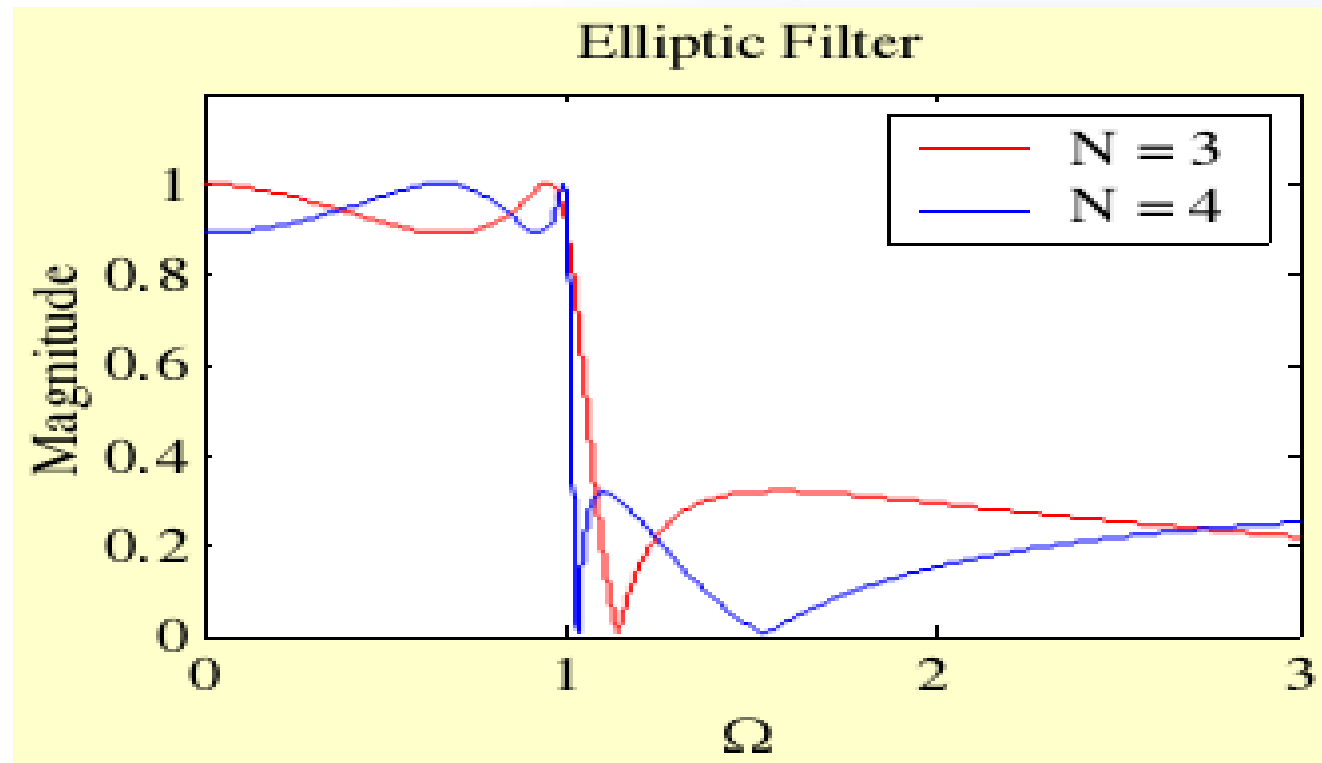
$$|H_a(j\Omega)|^2 = \frac{1}{1 + \varepsilon^2 R_N^2(\Omega / \Omega_p)}$$

where $R_N(\Omega)$ is a rational function of order N satisfying $R_N(1/\Omega) = 1/R_N(\Omega)$, with the roots of its numerator lying in the interval $0 < \Omega < 1$ and the roots of its denominator lying in the interval $1 < \Omega < \infty$.

Elliptic Approximation



Typical magnitude response plots are shown below:



Elliptic Approximation



For given Ω_p , Ω_s , ε , and A , the filter order can be estimated using:

$$N \cong \frac{2 \log_{10}(4 / k_1)}{\log_{10}(1 / \rho)}$$

where

$$k' = \sqrt{1 - k^2}$$

$$\rho_0 = \frac{1 - \sqrt{k'}}{2(1 + \sqrt{k'})}$$

$$\rho = \rho_0 + 2(\rho_0)^5 + 15(\rho_0)^9 + 150(\rho_0)^{13}$$

Elliptic Approximation



Example

- Determine the lowest order of a elliptic lowpass filter with a 1-dB cutoff frequency at 1 kHz and a minimum attenuation of 40 dB at 5 kHz.

Note: $k = 0.2$ and $1/k_1 = 196.5134$

Substituting these values we get:

$k' = 0.979796$, $\rho_0 = 0.00255135$, $\rho = 0.0025513525$

hence $N = 2.23308$

Choose $N = 3$.

A Comparison of the Analog Filter Types



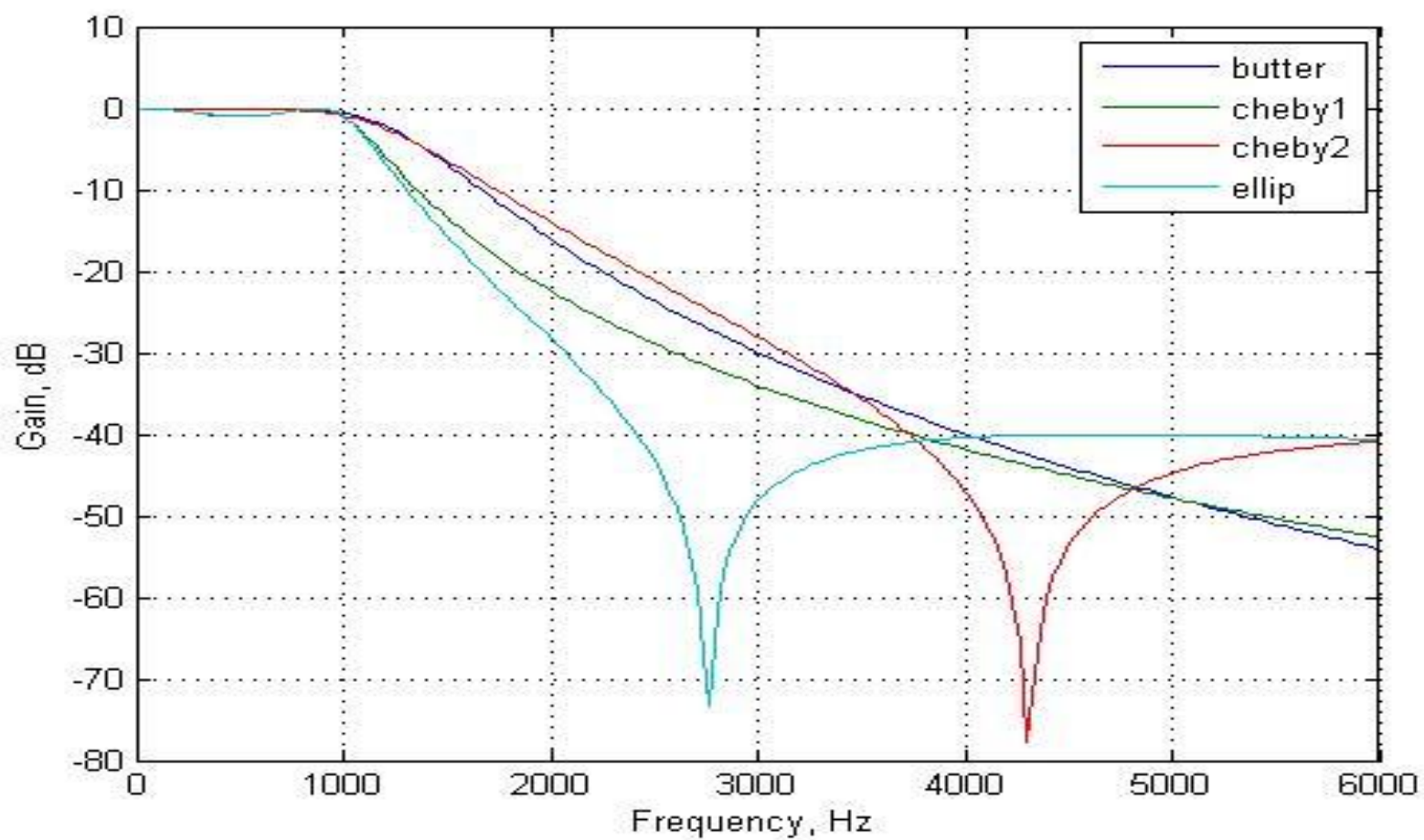
- The Butterworth filter has the widest transition band, with a **monotonically** decrease gain response.
- The elliptic filter has the **narrowest transition band**, with the equiripple passband and equiripple stopband response.
- The magnitude response of the Type2 Chebyshev filter in the passband is nearly identical to that of the Butterworth.
- The Butterworth and Chebyshev filters have a nearly linear phase response over about 3/4 of passband, whereas the elliptic filter has a nearly linear phase response over about 1/2 of the passband.

Analog Lowpass Filter Design Using MATLAB



Example: Design an analog lowpass filter of lowest order with a 1-dB cutoff frequency at 1kHz and a minimum attenuation of 40 dB at 4kHz.

```
[N,Wn] = buttord(Wp, Ws, Rp, Rs, 's');  
[num,den] = butter(N,Wn,'s');  
[N,Wn] = ellipord(Wp, Ws, Rp, Rs, 's');  
[num,den] = ellip(N, Rp, Rs, Wn, 's');  
[N,Wn] = cheb1ord(Wp, Ws, Rp, Rs, 's');  
[num,den] = cheby1(N,Rp,Wp,'s');  
[num,den] = cheby2(N,Rs,Ws,'s');  
omega = [0: 200: 12000*pi];  
h = freqs(num, den, omega);
```



9.2 The Design Methods of IIR Filter



- The core idea of design this kind of filter is to convert an analog transfer function $H_a(s)$ into a digital one $G(z)$.
- There are two common methods:
 - (1) The impulse invariance method
(Problem 9.6—P.520)
 - (2) The bilinear transformation method
(9.2—P.494)

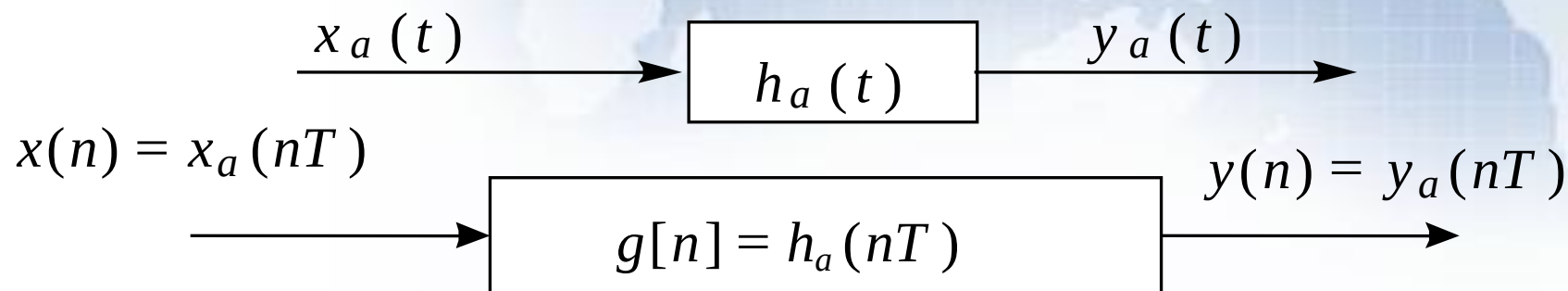
Supplement: The Impulse Invariance Method



- The Main Idea:

If a prototype causal analog transfer function $H_a(s)$ is wanted, and its impulse response is $h_a(t)$, the impulse response of the digital transfer function $G(z)$ satisfies:

$$g[n] = h_a(nT) \quad n = 0, 1, 2, \dots$$



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The Relationship Between ZT and LT



- **LT** - used in continuous signal analysis

ZT - used in discrete signal analysis

- **Given continuous signal $x_a(t)$, sampled signal $\hat{x}_a(t)$, their LT are:**

$$X_a(s) = L[x_a(t)]$$

$$\hat{X}_a(s) = L[\hat{x}_a(t)]$$

$$\because \hat{x}_a(t) = x_a(t) \big|_{t=nT} = \sum_{n=-\infty}^{\infty} x_a(nT) \delta(t - nT)$$

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$$\begin{aligned}\hat{X}_a(s) &= \int_{-\infty}^{\infty} \sum_{n=-\infty}^{\infty} x_a(nT) \delta(t - nT) e^{-st} dt \\ &= \sum_{n=-\infty}^{\infty} x_a(nT) \int_{-\infty}^{\infty} \delta(t - nT) e^{-st} dt \\ &= \sum_{n=-\infty}^{\infty} x_a(nT) e^{-nsT}\end{aligned}$$

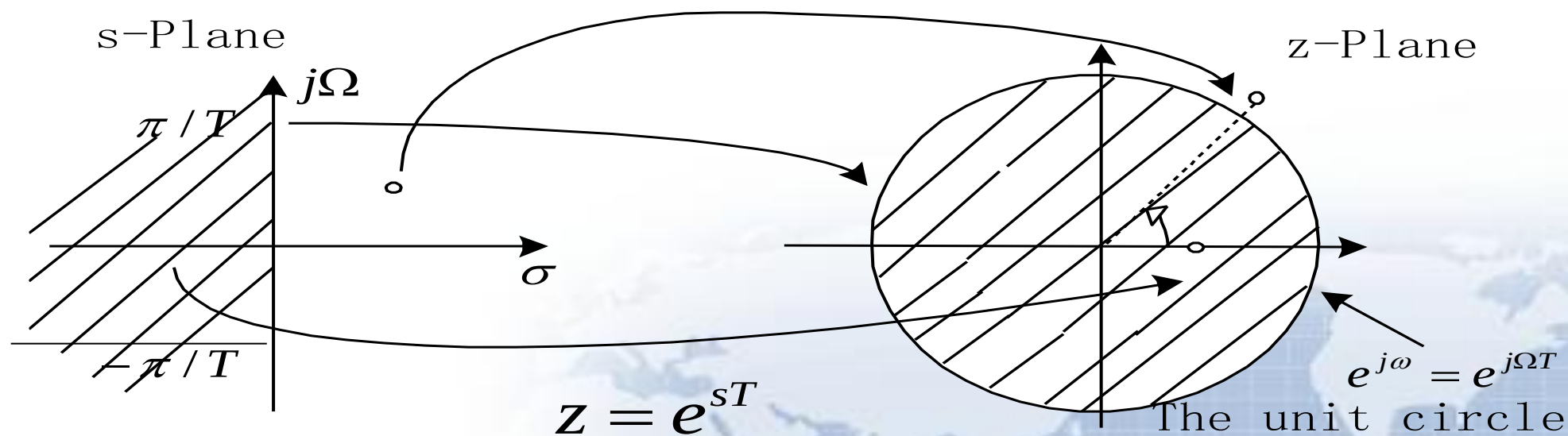
The ZT of the sampled sequence $x[n] = x_a(nT)$

$$X(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n}$$

So we can see: when $z = e^{sT}$, ZT=LT.

The relationship between them is a kind of mapping from s-plane to z-plane.

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if, $H_a(s) = L(h_a(t))$, $g[n] = h_a(nT)$,

$$G(z)|_{z=e^{sT}} = \frac{1}{T} \sum_{k=-\infty}^{\infty} H_a(s - jk \frac{2\pi}{T})$$

$$G(e^{j\omega}) = \frac{1}{T} \sum_{k=-\infty}^{\infty} H_a(j\Omega - jk \frac{2\pi}{T}) = \frac{1}{T} \sum_{k=-\infty}^{\infty} H_a(j \frac{\omega - k2\pi}{T})$$

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Discussion:



- The mapping satisfies the essential properties ($j\Omega$ -axis mapped onto the unit circle, stable region in s-plane mapped into the stable region in z-plane).
- The relation of the normalized angular frequency ω and the analog angular frequency Ω is $\omega = \Omega T$.
- when Nyquist theorem is satisfied,

$$G(e^{j\omega}) = \frac{1}{T} H_a(j\Omega), (|\omega| < \pi)$$

- If $H_a(j\Omega)$ is not band-limited, $G(e^{j\omega})$ obtained by this method is overlapped in frequency-domain.

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- **The Design Steps:**

(1) Expand $H_a(s)$ into partial-fractional:

$$H_a(s) = \sum_{k=1}^N \frac{A_k}{s - s_k} \quad (\text{Re}(s_k)_{\max} < 0)$$

(2) Based on $g(n) = h_a(nT)$, we can get:

$$G(z) = \sum_{k=1}^N \frac{A_k}{1 - e^{s_k T} z^{-1}}$$

Proof: $\because h_a(t) = L^{-1}[H_a(s)] = \sum_{k=1}^N A_k e^{s_k t} \mu(t) \therefore g[n] = h_a(nT) = \sum_{k=1}^N A_k e^{s_k nT} \mu[n]$

$$\Rightarrow G(z) = \sum_{n=-\infty}^{\infty} g[n] z^{-n} = \sum_{n=0}^{\infty} \sum_{k=1}^N A_k (e^{s_k T} z^{-1})^n = \sum_{k=1}^N A_k \sum_{n=0}^{\infty} (e^{s_k T} z^{-1})^n$$

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(3) Usually, we scale T to $G(z)$, and get:

$$G(z) = \sum_{k=1}^N \frac{TA_k}{1 - e^{s_k T} z^{-1}}$$

∴ From equation:

$$G(e^{j\omega}) = \frac{1}{T} H_a(j\Omega), \quad (|\omega| < \pi)$$

We can see the frequency response is inverse proportion of T . So we should amend it by:

$$g[n] = Th_a(nT)$$

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Discussion: $H_a(s)$ and $G(z)$



- The single pole s_k in s-plane mapping into the single pole $z=e^{skT}$ in z-plane.
- They both have the same coefficients A_k .
- If the analog filter is stable, that is $\text{Re}[s_k]<0$, then after conversion, the poles meet: $|e^{skT}|<1$, so the digital filter is stable.
- The impulse invariance method only ensure relationship of the poles between s and z-plane; but for zeros, they don't exist.

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Example



If $H_a(s) = \frac{2}{s^2 + 3s + 2}$, the sampling period is T .

Then: $H_a(s) = \frac{2}{s+1} - \frac{2}{s+2}$ $s_1 = -1, s_2 = -2$

Using the impulse invariance method, we get:

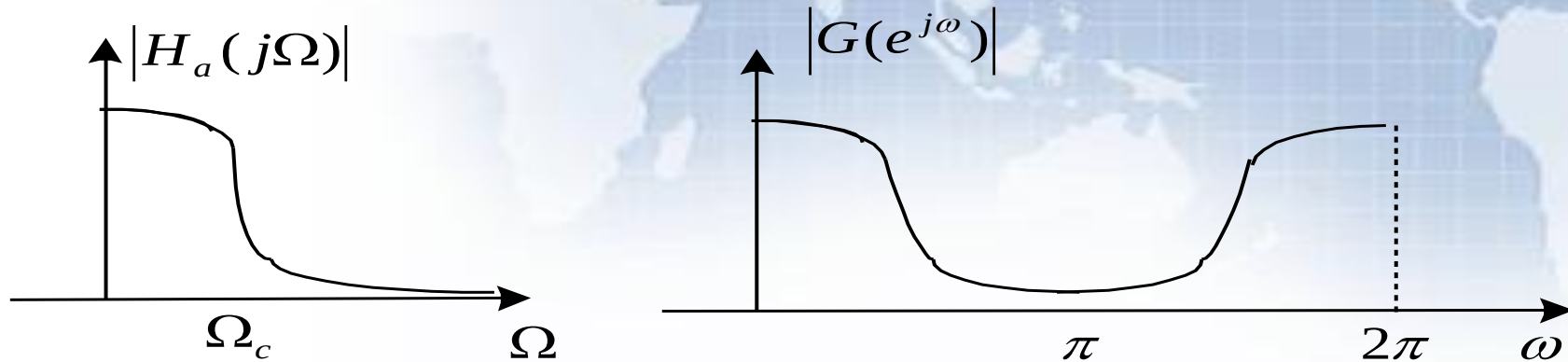
$$\begin{aligned} G(z) &= T \frac{2}{1 - e^{-T} z^{-1}} - T \frac{2}{1 - e^{-2T} z^{-1}} \\ &= \frac{2T(e^{-T} - e^{-2T})z^{-1}}{1 - (e^{-2T} + e^{-T})z^{-1} + e^{-3T}z^{-2}} \end{aligned}$$

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When $T=1$,

$$G(z) = \frac{0.4651z^{-1}}{1 - 0.5032z^{-1} + 0.04979z^{-2}}$$

Note: $G(e^{j\omega})$ is overlapped in the frequency-domain.



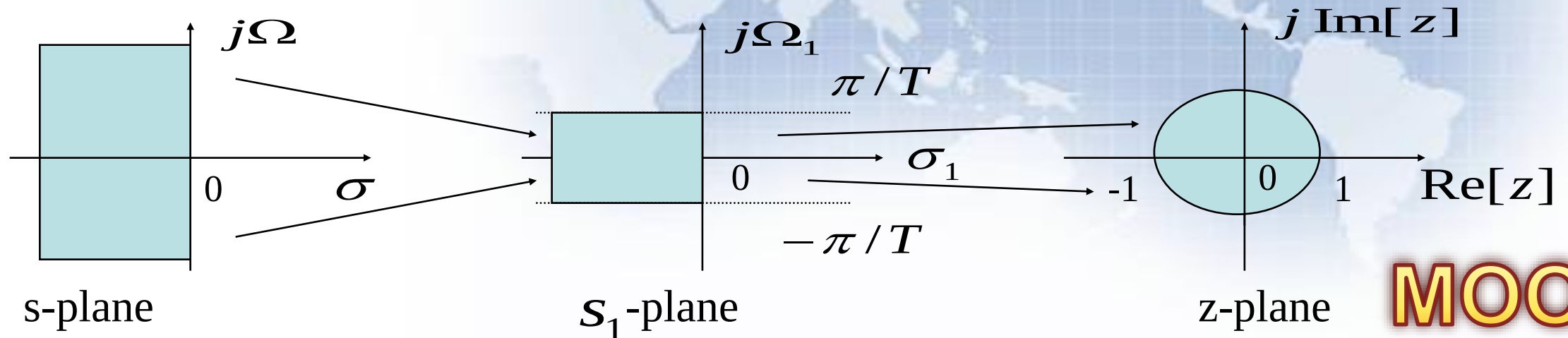
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9.2.1 The Bilinear Transformation



- Transformation theory:

For conquering the effect of multi-value mapping from s to z -plane, we want to find a new mapping:



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The Bilinear Transformation Method



- The bilinear transformation from s to z-plane:

$$s = \frac{2}{T} \frac{1 - z^{-1}}{1 + z^{-1}}$$

So, The relation between $G(z)$ and $H_a(s)$ is :

$$G(z) = H_a(s) \Big|_{s = \frac{2}{T} \frac{1 - z^{-1}}{1 + z^{-1}}}$$

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Note: the parameter T has no effect on the expression for $G(z)$, and we can choose $T=2$ to simplify the design procedure.

Discussion:



- Inverse bilinear transformation for $T=2$ is:

$$z = \frac{1+s}{1-s}$$

For $s = \sigma_0 + j\Omega_0$

$$z = \frac{(1+\sigma_0) + j\Omega_0}{(1-\sigma_0) - j\Omega_0} \Rightarrow |z|^2 = \frac{(1+\sigma_0)^2 + \Omega_0^2}{(1-\sigma_0)^2 + \Omega_0^2}$$

Thus, $\sigma_0 = 0 \rightarrow |z| = 1$

$\sigma_0 < 0 \rightarrow |z| < 1$

$\sigma_0 > 0 \rightarrow |z| > 1$

This mapping also satisfies the essential properties.

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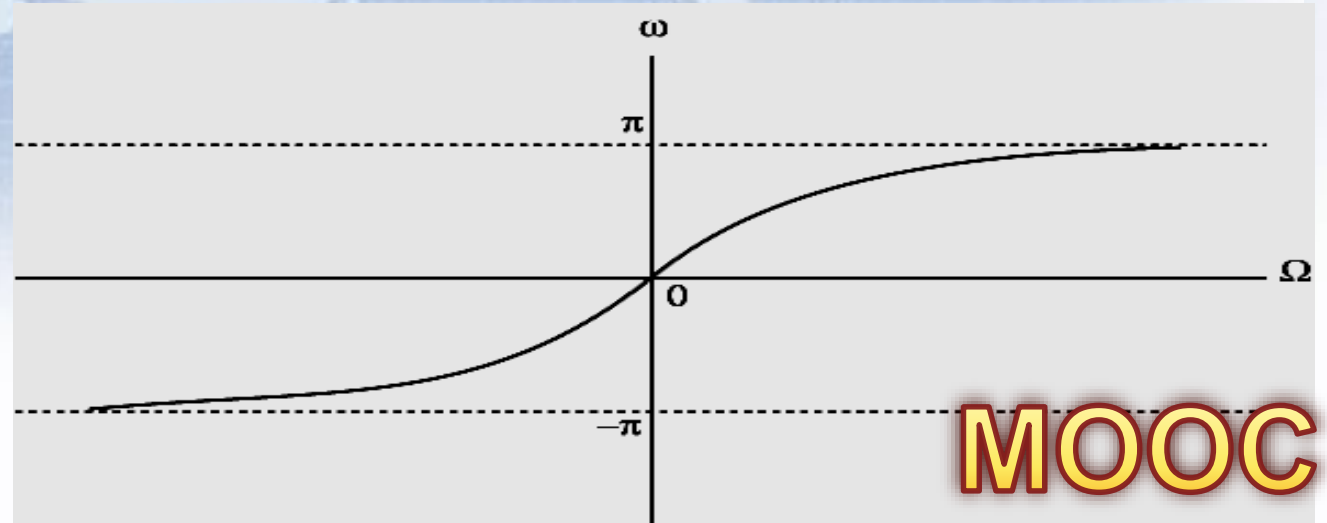
- For $z=e^{j\omega}$ with $T = 2$ we have:

$$j\Omega = \frac{1-e^{-j\omega}}{1+e^{-j\omega}} = \frac{e^{-j\omega/2}(e^{j\omega/2} - e^{-j\omega/2})}{e^{-j\omega/2}(e^{j\omega/2} + e^{-j\omega/2})}$$

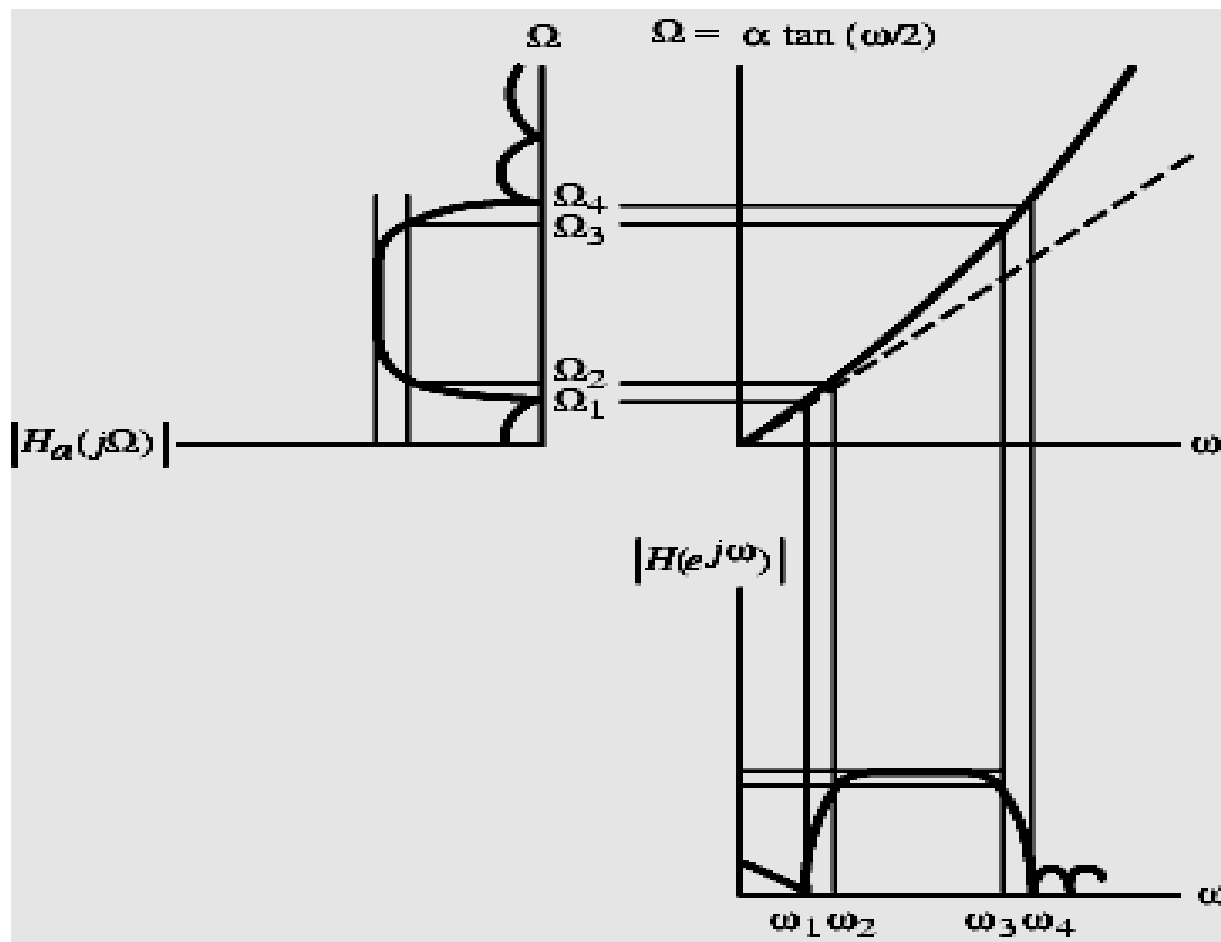
$$= \frac{j2\sin(\omega/2)}{2\cos(\omega/2)} = j\tan(\omega/2)$$

So, $\Omega = \tan(\omega/2)$

This introduces a distortion in frequency axis called frequency wrapping.



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Thus, we must first *prewarp* the critical bandedge frequencies ω_s and ω_p to find their analog equivalents Ω_p and Ω_s by using following equation:

$$\Omega = \tan(\omega/2)$$

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The design steps:



- (1) **Prewarp** (ω_p, ω_s) to find their analog equivalents (Ω_p, Ω_s);
 - (2) **Design the analog prototype filter** $H_a(s)$;
 - (3) **Obtain the digital filter** $G(z)$ by applying **bilinear transformation** to $H_a(s)$;
- **Bilinear transformation preserves the magnitude response of analog filter only if piecewise constant magnitude.**
 - **Bilinear transformation does not preserve phase response of analog filter.**

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9.2.2 Design of Low-Order Digital Filters



- Example – Consider:

$$H_a(s) = \frac{\Omega_c}{s + \Omega_c}$$

Applying bilinear transformation to the above we get the transfer function of a first-order digital lowpass Butterworth filter

$$G(z) = H_a(s) \Big|_{s=\frac{1-z^{-1}}{1+z^{-1}}} = \frac{\Omega_c (1+z^{-1})}{(1-z^{-1}) + \Omega_c (1+z^{-1})}$$

Design of Low-Order Digital Filters



- Rearranging terms we get:

$$G(z) = \frac{1-\alpha}{2} \cdot \frac{1+z^{-1}}{1-\alpha z^{-1}}$$

where

$$\alpha = \frac{1-\Omega_c}{1+\Omega_c} = \frac{1-\tan(\omega_c / 2)}{1+\tan(\omega_c / 2)}$$

Design of Low-Order Digital Filters



- Consider the second-order analog notch transfer function:

$$H_a(s) = \frac{s^2 + \Omega_o^2}{s^2 + Bs + \Omega_o^2}$$

where $|H_a(j\Omega_0)| = 0$ and $|H_a(j0)| = |H_a(j\infty)| = 1$

- Ω_0 is called the **notch frequency**
- If $|H_a(j\Omega_2)| = |H_a(j\Omega_1)| = 1/\sqrt{2}$ then
 $B = \Omega_2 - \Omega_1$ is the **3-dB notch bandwidth**



• Then $G(z) = H_a(s) \Big|_{s=\frac{1-z^{-1}}{1+z^{-1}}}$

$$= \frac{(1 + \Omega_o^2) - 2(1 - \Omega_o^2)z^{-1} + (1 + \Omega_o^2)z^{-2}}{(1 + \Omega_o^2 + B) - 2(1 - \Omega_o^2)z^{-1} + (1 + \Omega_o^2 - B)z^{-2}}$$
$$= \frac{1 + \alpha}{2} \cdot \frac{1 - 2\beta z^{-1} + z^{-2}}{1 - 2\beta(1 + \alpha)z^{-1} + \alpha z^{-2}}$$

where

$$\alpha = \frac{1 + \Omega_o^2 - B}{1 + \Omega_o^2 + B} = \frac{1 - \tan(B_\omega / 2)}{1 + \tan(B_\omega / 2)}$$

$$\beta = \frac{1 - \Omega_o^2}{1 + \Omega_o^2} = \frac{1 - \tan^2(\omega_o / 2)}{1 + \tan^2(\omega_o / 2)} = \cos \omega_o$$



- Example - Design a 2nd-order digital notch filter operating at a sampling rate of 400 Hz with a notch frequency at 60 Hz, 3-dB notch bandwidth of 6 Hz

Thus $\omega_0 = 2\pi(60/400) = 0.3 \pi$

$$B_w = 2\pi(6/400) = 0.03 \pi$$

From the above values we get:

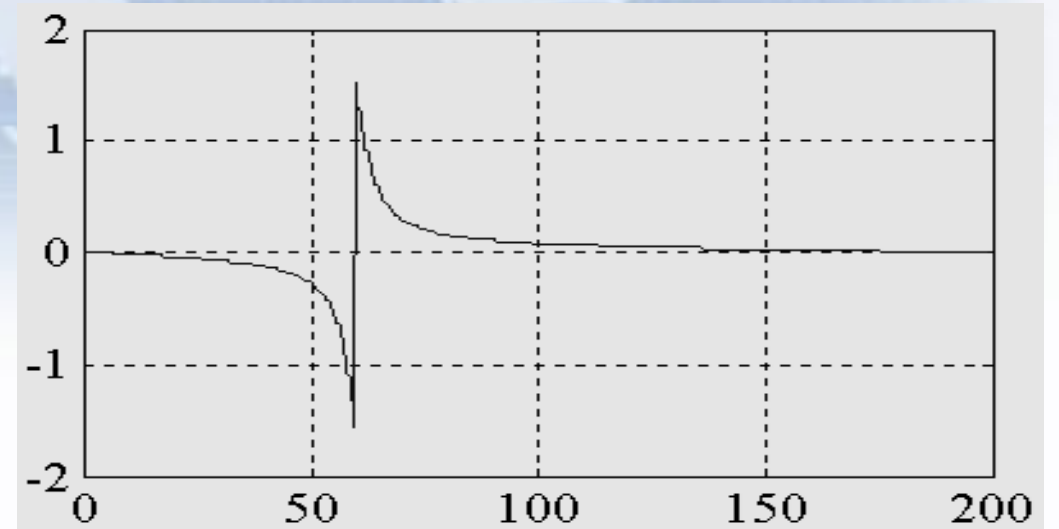
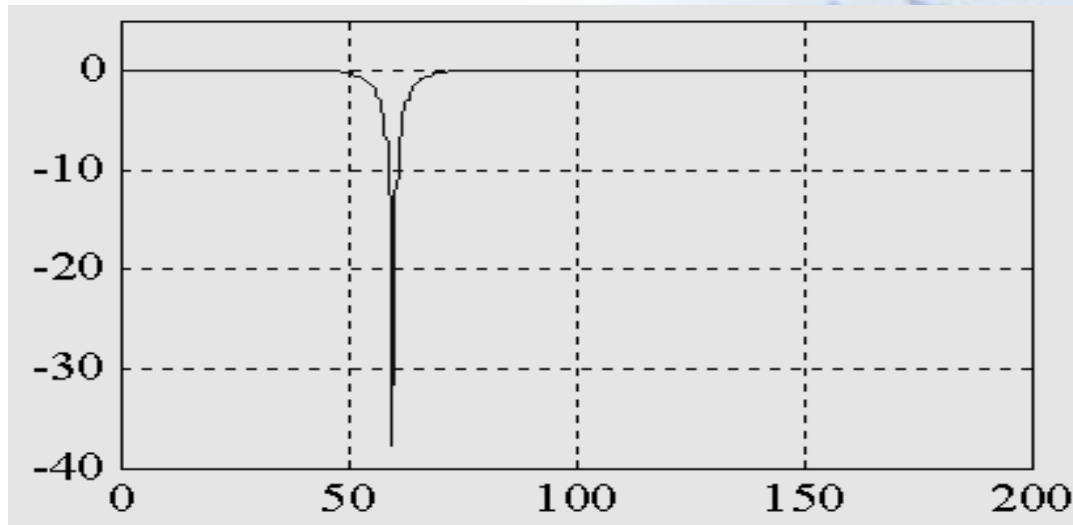
$$\alpha = 0.90993$$

$$\beta = 0.587785$$

Thus

$$G(z) = \frac{1+\alpha}{2} \cdot \frac{1-2\beta z^{-1} + z^{-2}}{1-2\beta(1+\alpha)z^{-1} + \alpha z^{-2}}$$
$$= \frac{0.954965 - 1.1226287 z^{-1} + 0.954965 z^{-2}}{1 - 1.1226287 z^{-1} + 0.90993 z^{-2}}$$

The gain and phase responses are shown below



9.3 Design of Lowpass IIR Digital Filters



Consider $G(z)$ with a maximally flat magnitude $\omega_p = 0.25\pi$, with a passband ripple not exceeding 0.5dB; $\omega_s = 0.55\pi$, with the minimum stopband attenuation at the stopband edge frequency ω_s is 15dB.

Base on this requirement, if $|G(e^{j0})| = 1$ we get:

$$20\log_{10}|G(e^{j0.25\pi})| \geq -0.5\text{dB}, \alpha_{\max} = 0.5\text{dB}$$

$$20\log_{10}|G(e^{j0.55\pi})| \leq -15\text{dB}, \alpha_s = 15\text{dB}$$

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1. The Bilinear Transformation Method



(1) Prewarpping:

$$\Omega_p = \tan\left(\frac{\omega_p}{2}\right) = \tan\left(\frac{0.25\pi}{2}\right) = 0.4142136$$

$$\Omega_s = \tan\left(\frac{\omega_s}{2}\right) = \tan\left(\frac{0.55\pi}{2}\right) = 1.1708496$$

(2) Design the prototype analog filter $H_a(s)$

passband ripple 0.5dB

$$\Rightarrow \varepsilon^2 = 0.1220185$$

stopband attenuation 15dB

$$\Rightarrow A^2 = 31.622777$$

$$k = \frac{\Omega_s}{\Omega_p} = \frac{1.1708496}{0.4142136} = 2.8266809$$

$$k_1 = \frac{\varepsilon}{\sqrt{A^2 - 1}} = 15.841979$$

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$$N = \frac{\log_{10}(1/k_1)}{\log_{10}(1/k)} = \frac{\log_{10}(15.841979)}{\log_{10}(2.8266814)} = 2.6586997$$

The least order of Butterworth LPF is: N=3

Using Eq.(A.8), we get the 3-dB cutoff frequency:

$$\Omega_c = (\varepsilon)^{-1/N} \Omega_p = 1.419915(\Omega_p) = 0.588148$$

The third-order normalized lowpass Butterworth transfer function with $\Omega_c=1$ is:

$$H_{an}(s) = \frac{1}{(s+1)(s^2+s+1)} = \frac{1}{s^3+2s^2+2s+1}$$

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Then the denormalized lowpass Butterworth TF is:

$$H_a(s) = H_{an}\left(\frac{s}{\Omega_c}\right) = H_{an}\left(\frac{s}{0.588148}\right)$$

(3) Design the digital filter $G(z)$ by applying bilinear transformation to $H_a(s)$:

$$\begin{aligned} G(z) &= H_a(s) \Big|_{s=\frac{1-z^{-1}}{1+z^{-1}}} \\ &= \frac{0.0662272(1+z^{-1})^3}{(1-0.2593284z^{-1})(1-0.6762858z^{-1}+0.3917468z^{-2})} \end{aligned}$$

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2. The Impulse Invariance Method



The third-order lowpass Butterworth transfer function which has 3-dB frequency at Ω_c is:

$$H_a(s) = H_{an}\left(\frac{s}{\Omega_c}\right) = \frac{1}{1 + 2\frac{s}{\Omega_c} + 2\left(\frac{s}{\Omega_c}\right)^2 + \left(\frac{s}{\Omega_c}\right)^3}$$

We can get the poles and represent it as

$$H_a(s) = \frac{\Omega_c^3}{(s + \Omega_c)(s - \Omega_c e^{j\frac{2\pi}{3}})(s - \Omega_c e^{-j\frac{2\pi}{3}})}$$



It is partial-fractional expressed as:

$$H_a(s) = \frac{\Omega_c}{s + \Omega_c} + \frac{-(\Omega_c / \sqrt{3})e^{j\pi/6}}{s + \Omega_c(1 - j\sqrt{3})/2} + \frac{-(\Omega_c / \sqrt{3})e^{-j\pi/6}}{s + \Omega_c(1 + j\sqrt{3})/2}$$

Then we get digital transfer function as:

$$G(z) = \frac{\omega_c}{1 - e^{-\omega_c} z^{-1}} + \frac{-(\omega_c / \sqrt{3})e^{j\pi/6}}{1 - e^{-\omega_c(1 - j\sqrt{3})/2} z^{-1}} + \frac{-(\omega_c / \sqrt{3})e^{-j\pi/6}}{1 - e^{-\omega_c(1 + j\sqrt{3})/2} z^{-1}}$$

$$\omega_c = \pi / 4$$

$$G(z) = \frac{0.785398}{1 - 0.455938z^{-1}} + \frac{-0.785398 + 0.604884z^{-1}}{1 - 1.0500756z^{-1} + 0.455938z^{-2}}$$

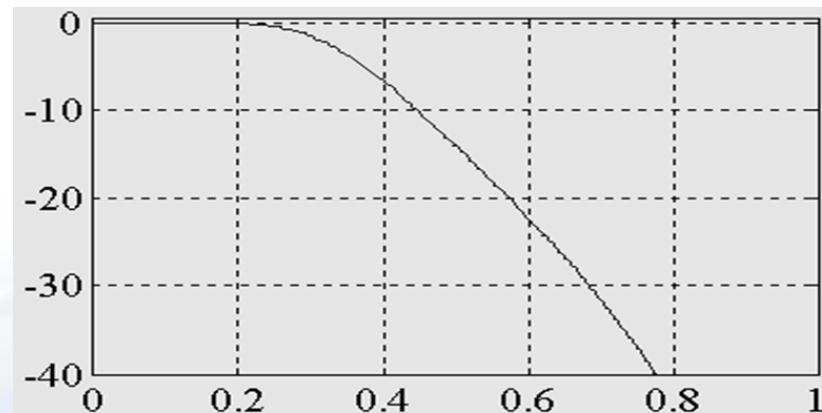
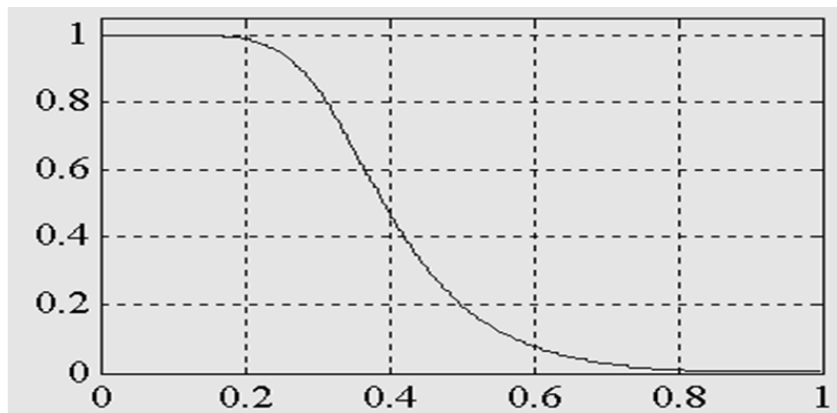


Finally,

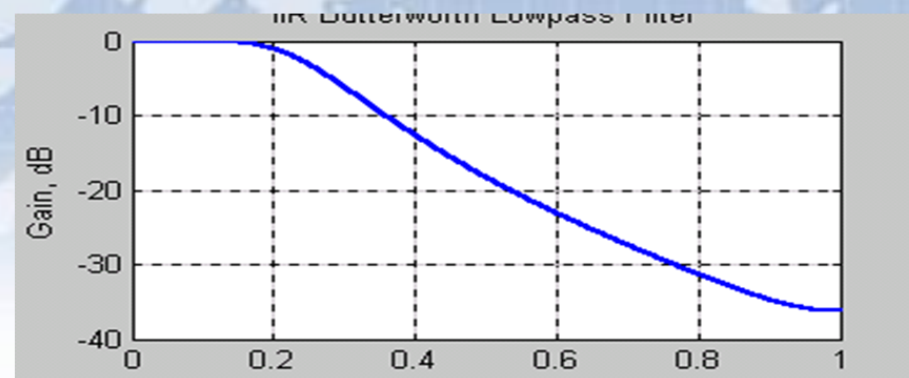
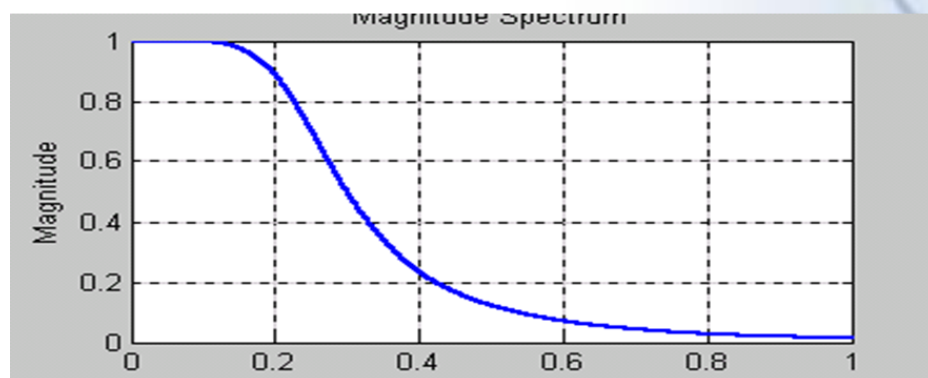
$$G(z) = \frac{0.13825z^{-1} + 0.08230z^{-2}}{1 - 1.50601z^{-1} + 0.93471z^{-2} - 0.20788z^{-3}}$$

Discussion:

Magnitude and gain responses of $G(z)$ derived by The Impulse Invariance Method and The Bilinear Translation Method are shown below:



The Bilinear Translation Method



The Impulse Invariance Method



Ex: Use a IIR digital filter to implement a Butterworth 3rd-order LPF with sampling interval $T = 250\mu s$, cutoff frequency: $f_c = 1 kHz$.

① Determine normalized frequency

$$\omega = \Omega T = \Omega \times 250 \times 10^{-6} = \frac{\Omega}{4} \times 10^{-3}$$

② Determine normalized cutoff frequency

$$\omega_c = \frac{2\pi f_c}{4} \times 10^{-3} = \frac{2\pi \times 10^3}{4} \times 10^{-3} = \frac{\pi}{2}$$

③ Prewarping frequency to find the equivalent analog cutoff frequency

$$\Omega_c = \tan \frac{\omega_c}{2} = \tan \frac{\pi}{4} = 1$$

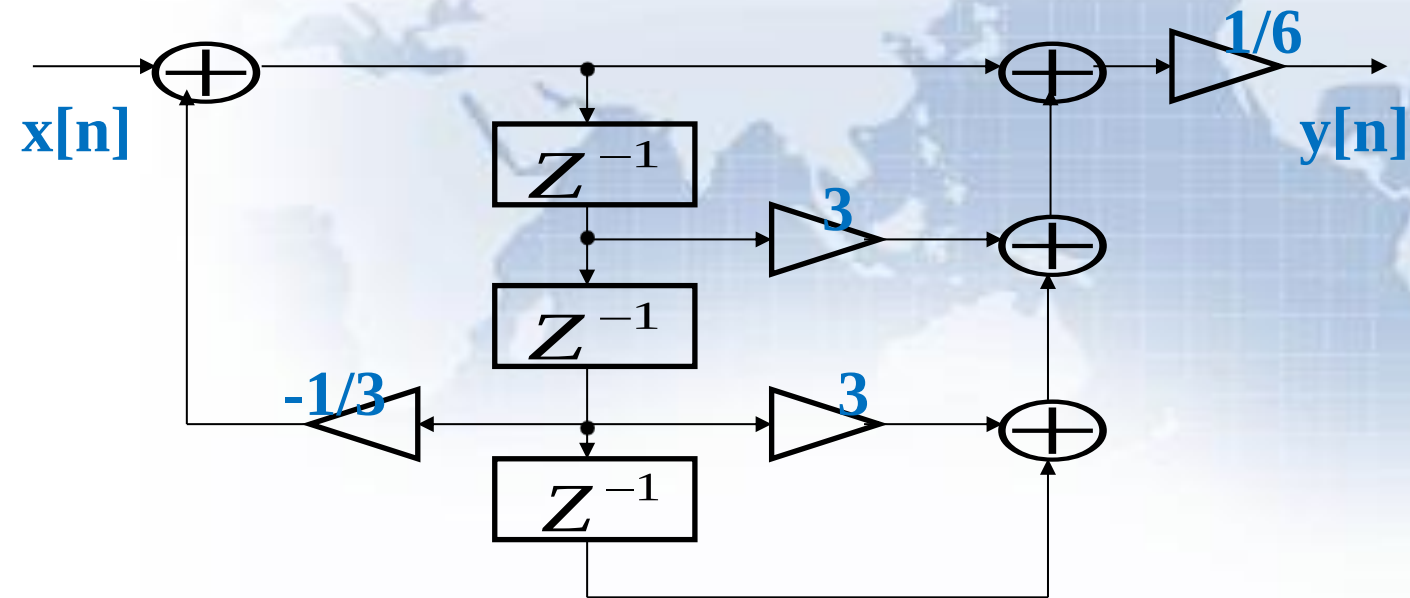
④ Choose the analog prototype filter

$$H_a(S) = \frac{1}{\left(\frac{S}{\Omega_c}\right)^3 + 2\left(\frac{S}{\Omega_c}\right)^2 + 2\left(\frac{S}{\Omega_c}\right) + 1} = \frac{1}{S^3 + 2S^2 + 2S + 1}$$

⑤ Determine the transfer function of the desired digital filter using the bilinear transformation

$$G(Z) = H_a(S) \Big|_{s=\frac{1-Z^{-1}}{1+Z^{-1}}} = \frac{1}{6} \cdot \frac{1+3Z^{-1}+3Z^{-2}+Z^{-3}}{1+\frac{1}{3}Z^{-2}}$$

⑥ Choose filter structure



9.4 Design of Highpass, Bandpass and Bandstop IIR Digital Filters



There are two approaches as follows:

- (1) Design other type analog filters firstly, then transform it into digital ones : Step1→Step5
 - (2) Design digital lowpass filter firstly, then transform it into other type digital filters: Step1→Step5
- Please look at P.501 to learn the detailed design procedure.

9.5 Spectral Transformations of IIR filters



Solving: to modify the characteristics of a filter to meet the new specifications without repeating the filter design procedure.

There are several conditions:

(1) Lowpass-to-lowpass transformation.

(2) Other transformation.

Please look at P.509 Table 9.1 to learn the detailed design procedure.

9.6 IIR Digital Filter Design Using MATLAB



- **Order Estimation:**
buttord, cheb1ord, cheb2ord, ellipord
- **Filter Design:**
butter, cheby1, cheby2, ellip
- **Filter Test: freqz**
- **Spectral Transformations:**
iirlp2lp, iirlp2hp, iirlp2bp, iirlp2bs
- **bilinear**



(1) Determine filter order N , frequency scaling factor W_n .

$[N, W_n] = \text{buttord} (W_p, W_s, R_p, R_s)$

$[N, W_n] = \text{cheb1ord} (W_p, W_s, R_p, R_s)$

$[N, W_n] = \text{cheb2ord} (W_p, W_s, R_p, R_s)$

$[N, W_n] = \text{ellipord} (W_p, W_s, R_p, R_s)$



(2) Determine the coefficients of the transfer function.

[b, a] = butter (N, Wn)

[b, a] = cheby1 (N, Rp, Wn)

[b, a] = cheby2 (N, Rs, Wn)

[b, a] = ellip (N, Rp, Rs, Wn)

Example 9.3



- Design of a Type 1 Chebyshev IIR digital highpass filter, Specifications:

$$F_p = 700\text{Hz}, F_s = 500\text{Hz},$$

$$\alpha_p = 1\text{ dB}, \alpha_s = 32\text{dB}, F_T = 2\text{ kHz}$$

- Normalized angular bandedge frequencies

$$\omega_p = 2\pi F_p / F_T = 2\pi \times 700 / 2000 = 0.7\pi$$

$$\omega_s = 2\pi F_s / F_T = 2\pi \times 500 / 2000 = 0.5\pi$$



Prewarping these frequencies we get

$$\hat{\Omega}_p = \tan(\omega_p / 2) = 1.9626105$$

$$\hat{\Omega}_s = \tan(\omega_s / 2) = 1.0$$

For the prototype analog lowpass filter choose $\Omega_p = 1$

Using $\Omega = -\frac{\Omega_p \hat{\Omega}_p}{\hat{\Omega}}$, we get $\Omega_s = 1.9626105$

• Analog lowpass filter specifications:

$$\Omega_p = 1, \Omega_s = 1.926105, \alpha_p = 1 \text{ dB}, \alpha_s = 32 \text{ dB}$$



- MATLAB code fragments used for the design

```
[N, Wn] = cheb1ord(1, 1.9626105, 1, 32, 's')
```

```
[B, A] = cheby1(N, 1, Wn, 's');
```

```
[BT, AT] = lp2hp(B, A, 1.9626105);
```

```
[num, den] = bilinear(BT, AT, 0.5);
```

- Using cheby1 design HP filter directly:

```
Wp=0.7; Ws=0.5;
```

```
Rp=1; Rs=32;
```

```
[N,Wn] = cheb1ord(Wp, Ws, Rp, Rs);
```

```
[b,a] = cheby1(N, Rp, Wn, 'high');
```



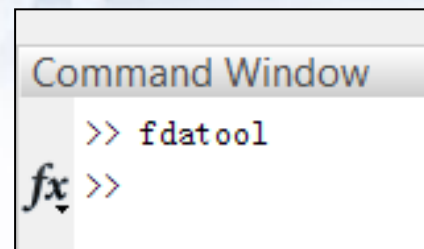
Digital Filter Design Using FDATool

- *Digital Filter Design Specifications*
- *Introduction to FDATool*
- *FIR & IIR Filter Design with FDATool*

Introduction to FDATool



- FDATool is a MATLAB toolbox that helps design digital filters in a visual manner
- FDATool can be used to design various filters, such as
 - FIR low-pass, high-pass, band-pass and band-stop filters
 - IIR low-pass, high-pass, band-pass and band-stop filters
- Type “fdatool” in the MATLAB command window to start FDATool



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Filter Design & Analysis Tool - [untitled.fda *]

File Edit Analysis Targets View Window Help

Current Filter Information

Structure: Direct-Form FIR
Order: 50
Stable: Yes
Source: Designed

Store Filter ...
Filter Manager ...

Filter Specifications

Mag. (dB)

0

ω_{pass} ω_{stop} 1 ω (normalized)

A_{pass}
 A_{stop}

Response Type

☒ Lowpass
☐ Highpass
☐ Bandpass
☐ Bandstop
☐ Differentiator

Design Method

☐ IIR Butterworth
☒ FIR Equiripple

Filter Order

☐ Specify order: 10
☒ Minimum order

Options

Density Factor: 20

Frequency Specifications

Units: Normalized (0 to 1)
Fs: 48000
wpass: 0.2
wstop: 0.3

Magnitude Specifications

Units: dB
Apass: 1
Astop: 80

Design Filter

Ready

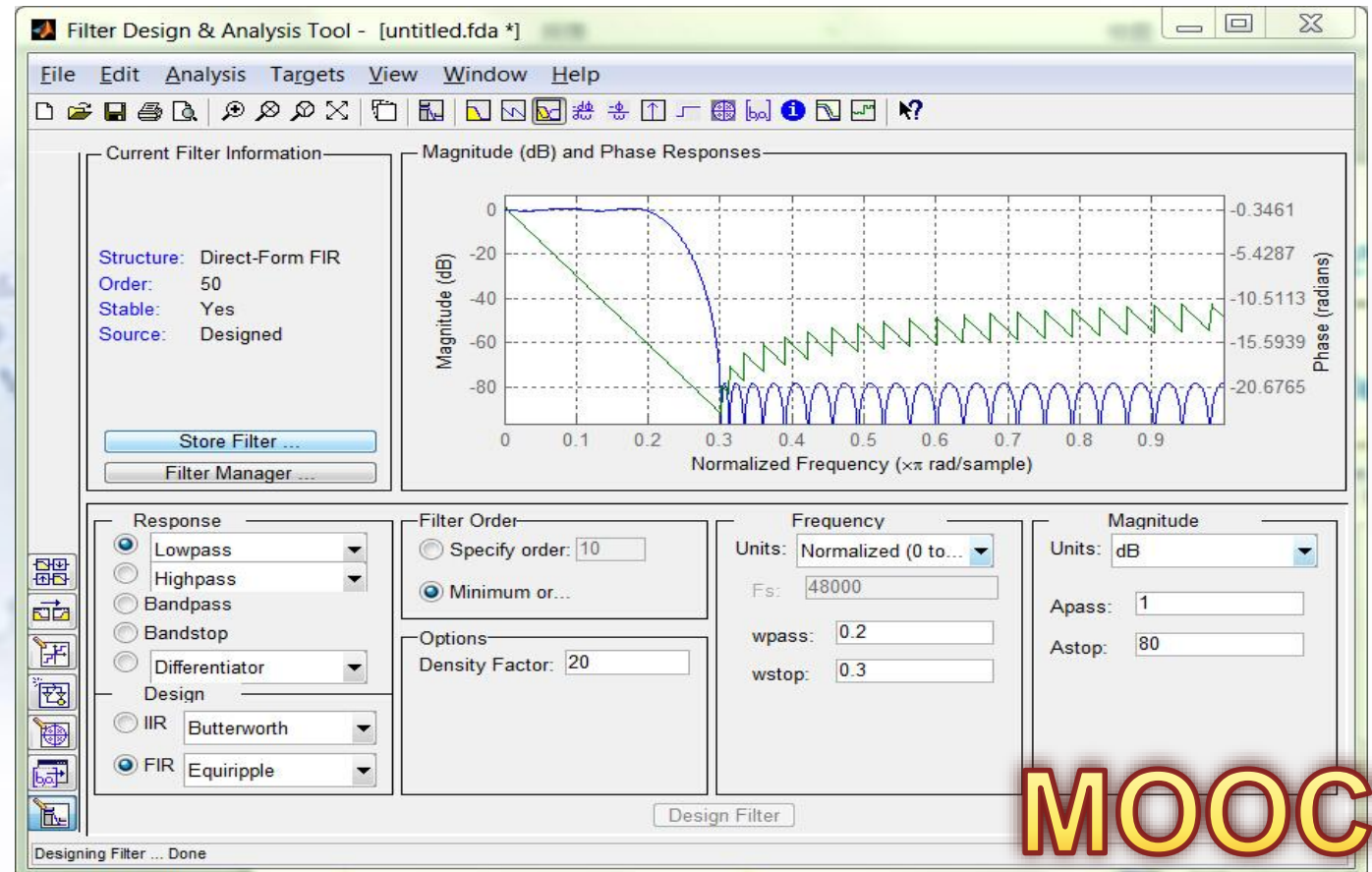
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FIR Digital Filter Design



- Example 1: Design an **FIR lowpass** filter with the specifications

- $\omega_p = 0.2\pi$, $\omega_s = 0.3\pi$,
- Max. passband ripple= 1dB
- Min. stopband attenuation=80dB



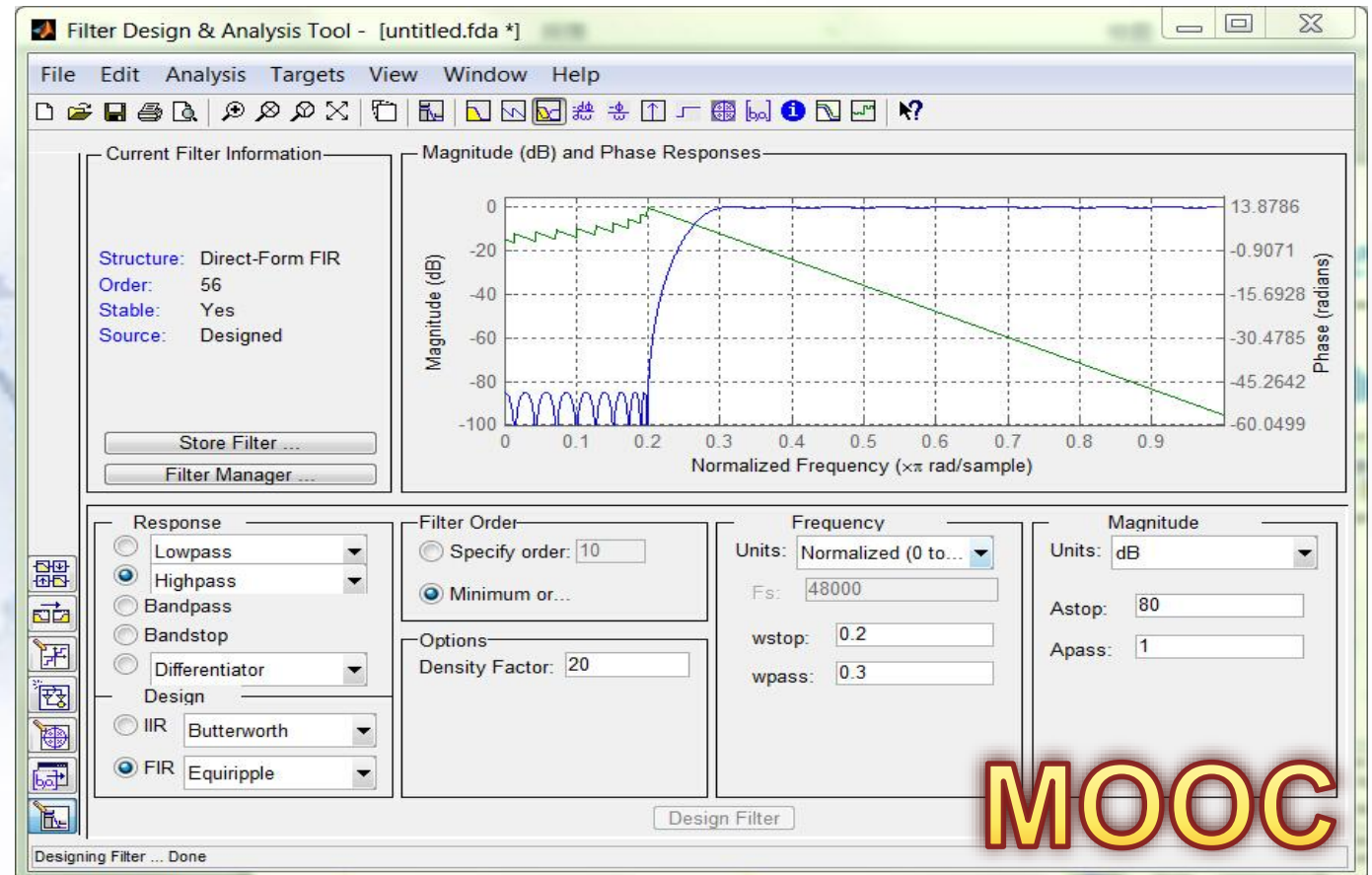
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FIR Digital Filter Design



- Example 2: Design an **FIR highpass** filter with the specifications

- $\omega_p = 0.3\pi$, $\omega_s = 0.2\pi$
- Max. passband ripple= 1dB
- Min. stopband attenuation=80dB



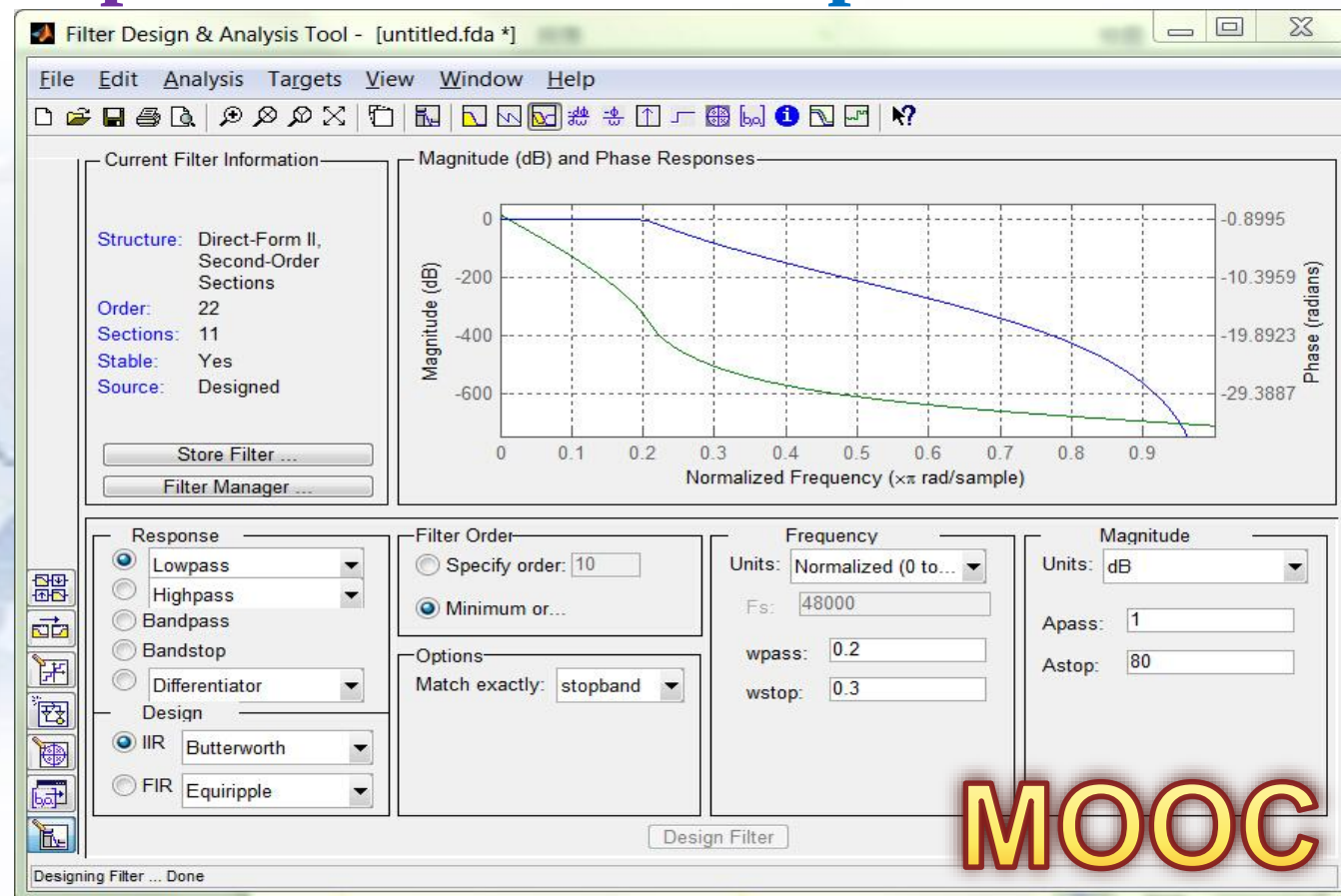
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IIR Digital Filter Design



- Example 3: Design an IIR lowpass filter with the specifications

- $\omega_p = 0.2\pi$, $\omega_s = 0.3\pi$
- Max. passband ripple= 1dB
- Min. stopband attenuation=80dB



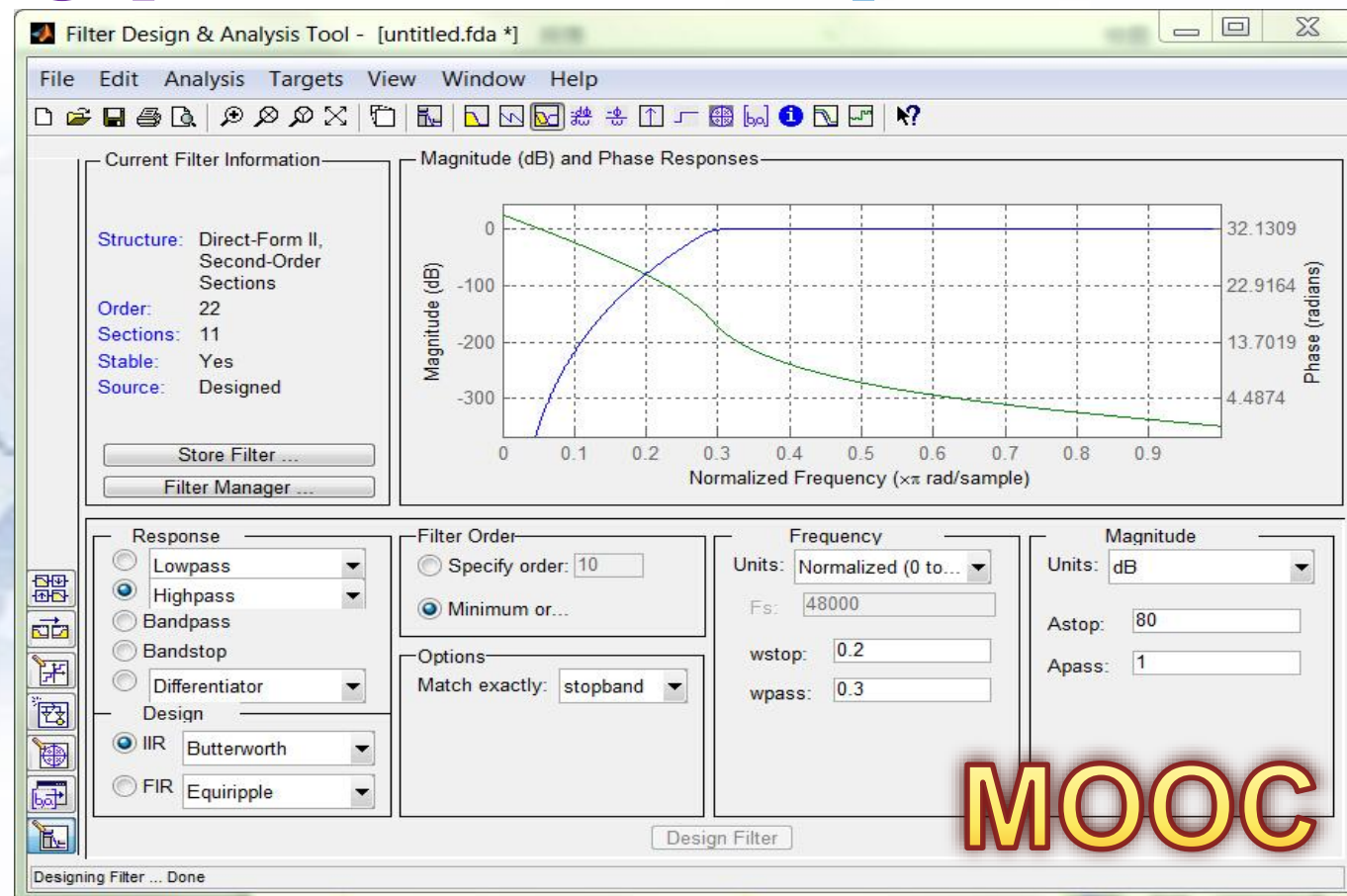
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IIR Digital Filter Design



- Example 4: Design an IIR highpass filter with the specifications

- $\omega_p = 0.3\pi$, $\omega_s = 0.2\pi$
- Max. passband ripple= 1dB
- Min. stopband attenuation=80dB



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Homework:



9.2, 9.4, 9.9(a), 9.10(a), 9.11(a), 9.12, 9.14, 9.16, 9.30
M9.1, M9.5, M9.8